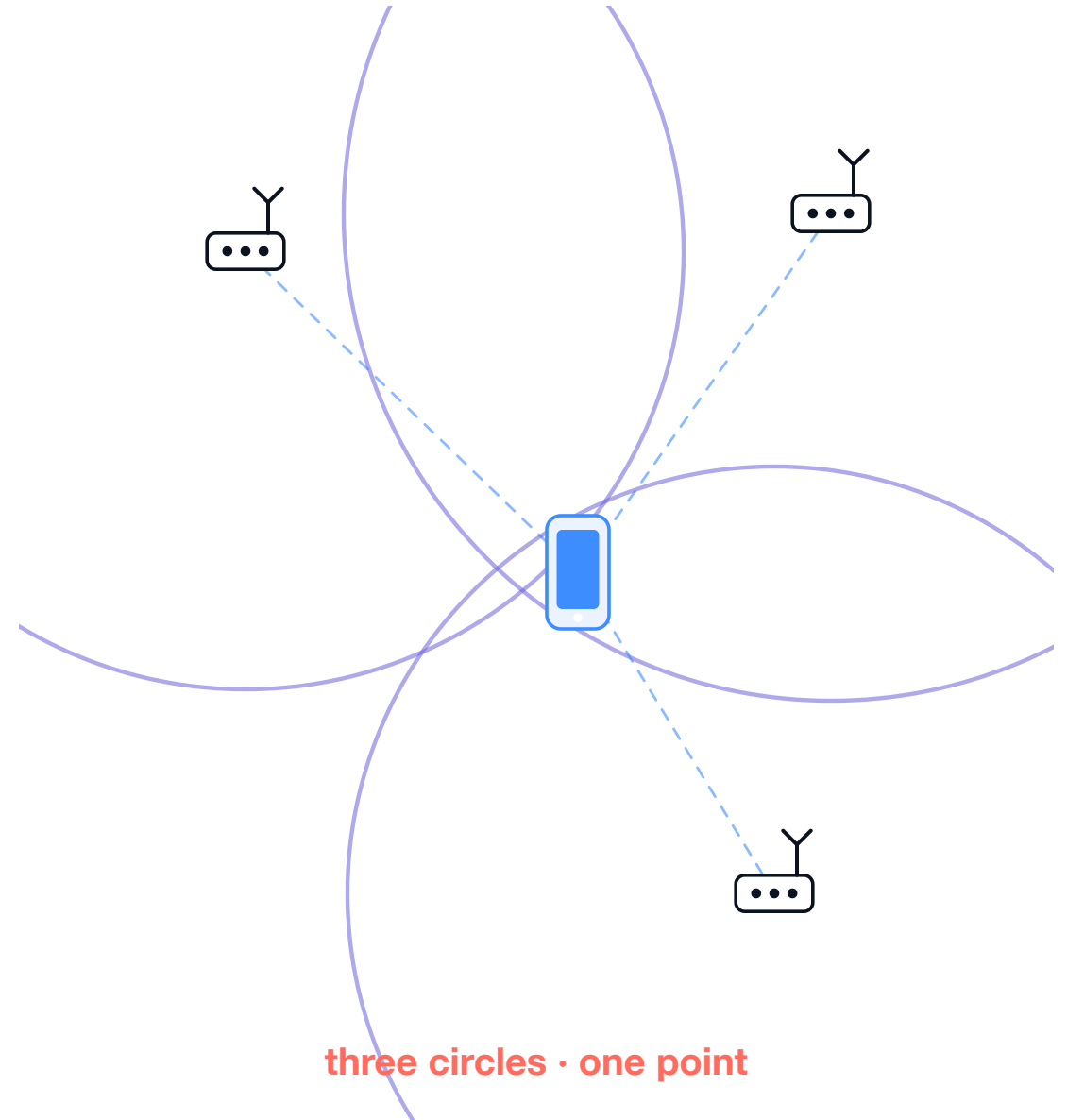


Wireless localization

A phone hears a signal from a radio at a known position. How strong it is, when it arrived and which way it came from each say something about where the phone is. This lecture turns those three measurements into a point on a floor plan, and asks what stops each of them from being exact.

Yaxiong Xie



Why does a phone need to know where it is?



turn-by-turn navigation

the map has to know which corner you are on



finding a lost key

a tag on the key, a phone that asks where it is



a robot in a warehouse

every shelf it drives to is a coordinate



a controller in the hand

the hand to the centimetre, all the time



an emergency call

the dispatcher needs the caller on a map



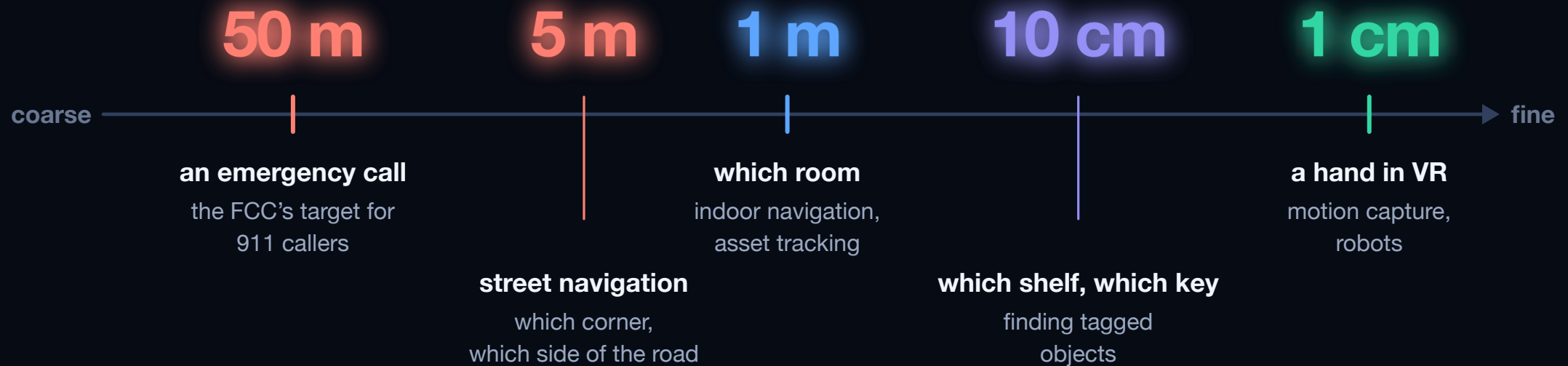
people and things nearby

which friend, which bus, which shared bike

Almost every mobile application starts by asking where the device is.

How accurate does each of them need to be?

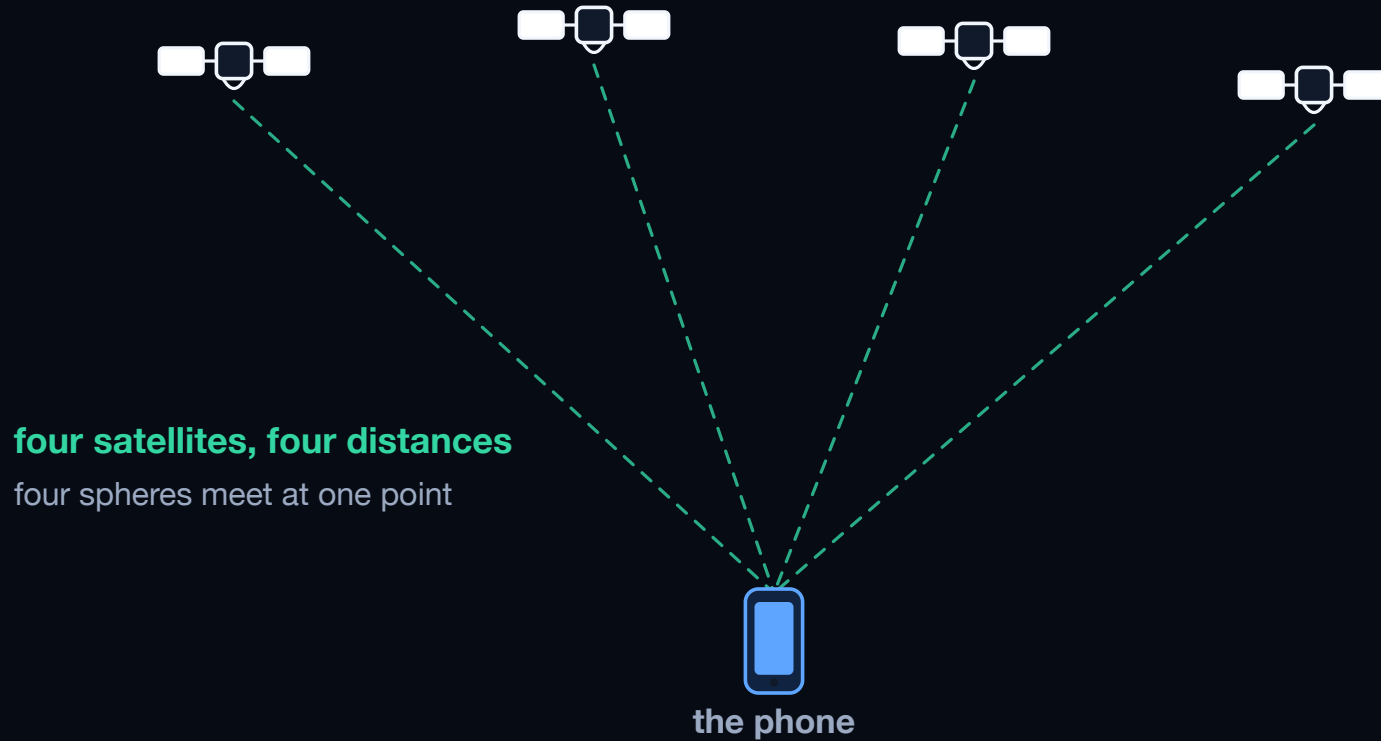
ONE AXIS, FROM A CITY BLOCK TO A FINGER WIDTH



Four orders of magnitude. No single method covers the whole axis.

Location is not one problem: the accuracy required spans a factor of 5,000.

Outdoors this is solved



4.9 m

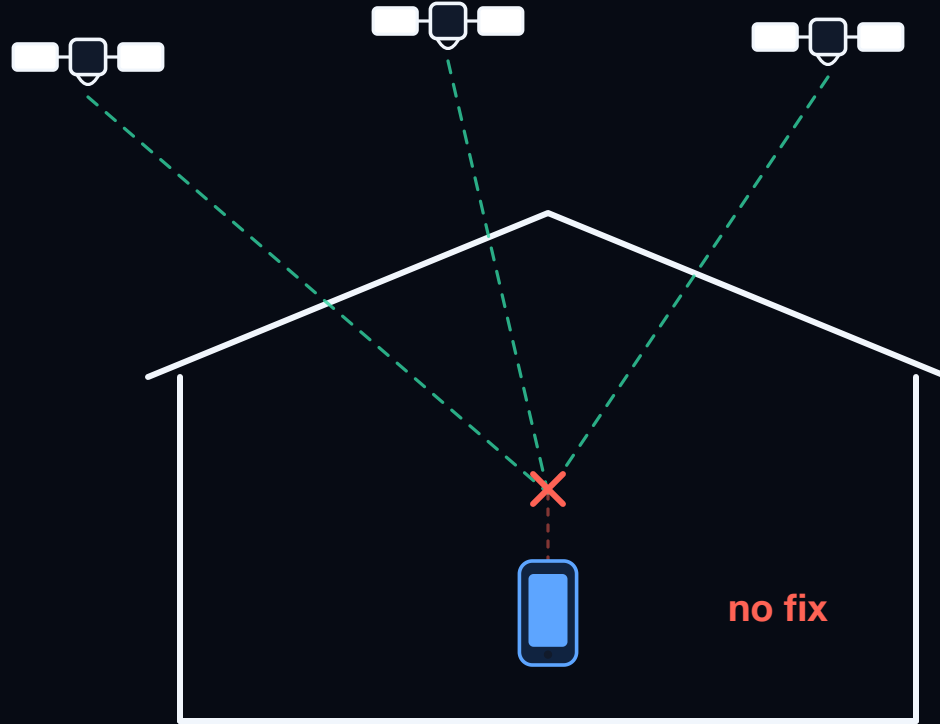
a phone under open sky

typical horizontal accuracy, as reported on gps.gov

20,000 km up, and the position comes out to 5 m.

GPS gives a phone about 5 m under open sky: the coarse end of the axis.

What happens to GPS indoors?



-130 dBm

the satellite signal at the ground

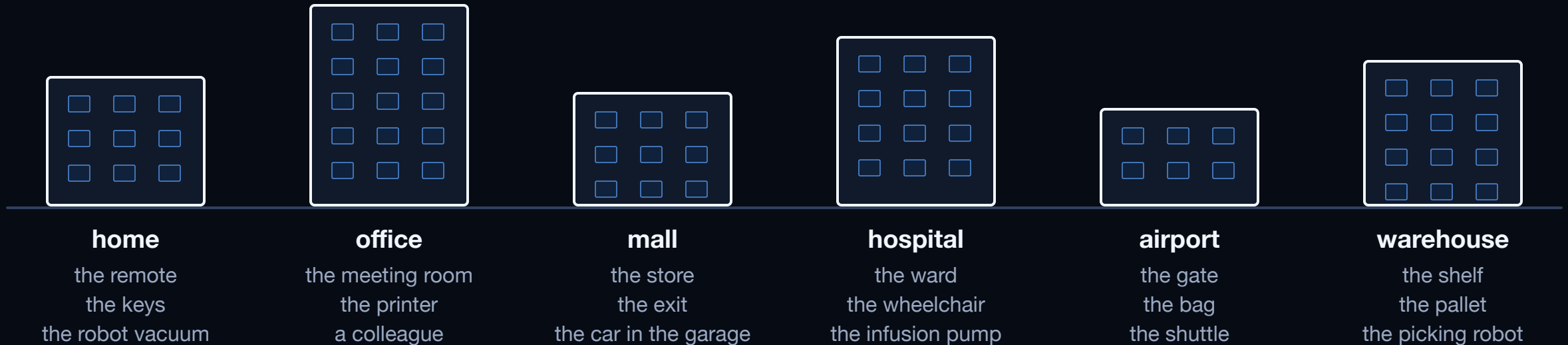
a tenth of a millionth of a millionth of a milliwatt

A roof and a few walls take another 10 to 30 dB.
Below that, the receiver cannot lock onto the code.

The signal that was barely enough outside is gone once there is a roof.

And indoors is where the phone spends its day

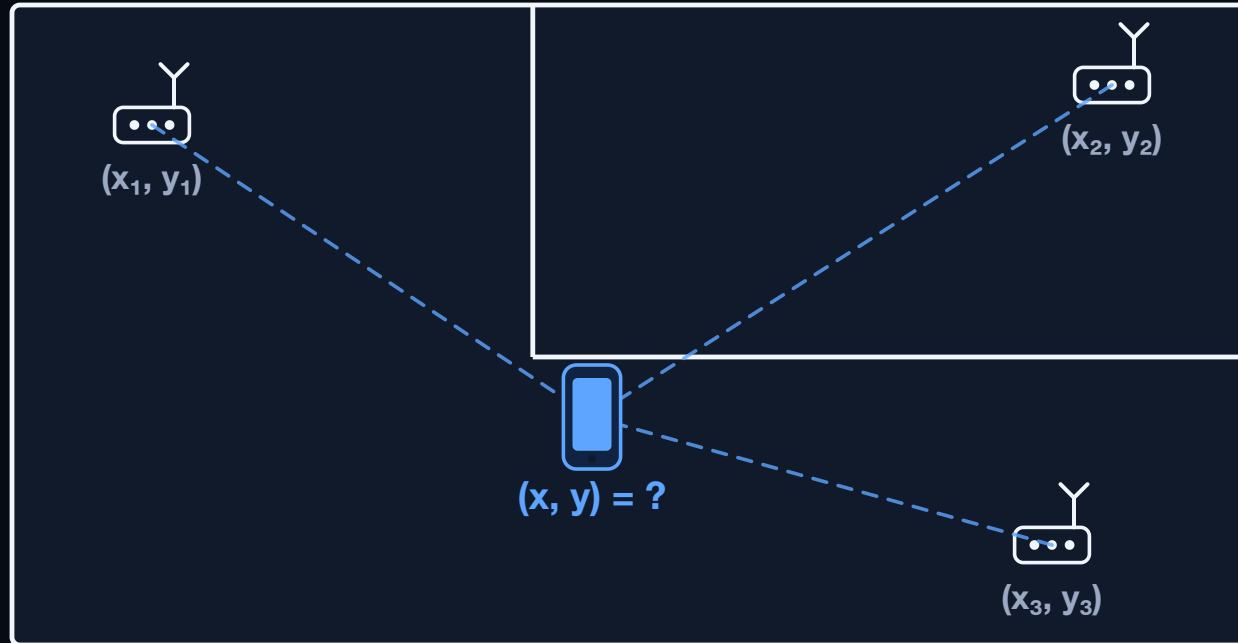
THE PLACES WHERE THE PHONE SPENDS ITS DAY



No satellite reaches any of these rooms. Every one of them has Wi-Fi.

Most of the things people want to find are in rooms that GPS cannot see.

What does “localization” mean?



known

where each anchor is: (x_i, y_i)

measured

one property of the signal from anchor i

unknown

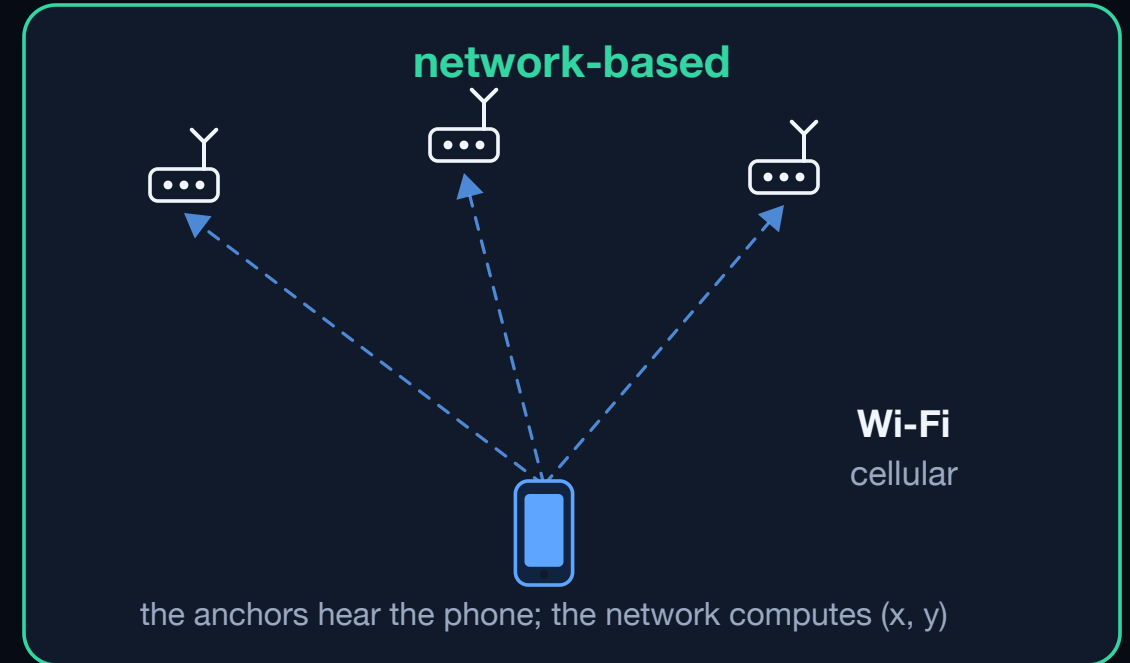
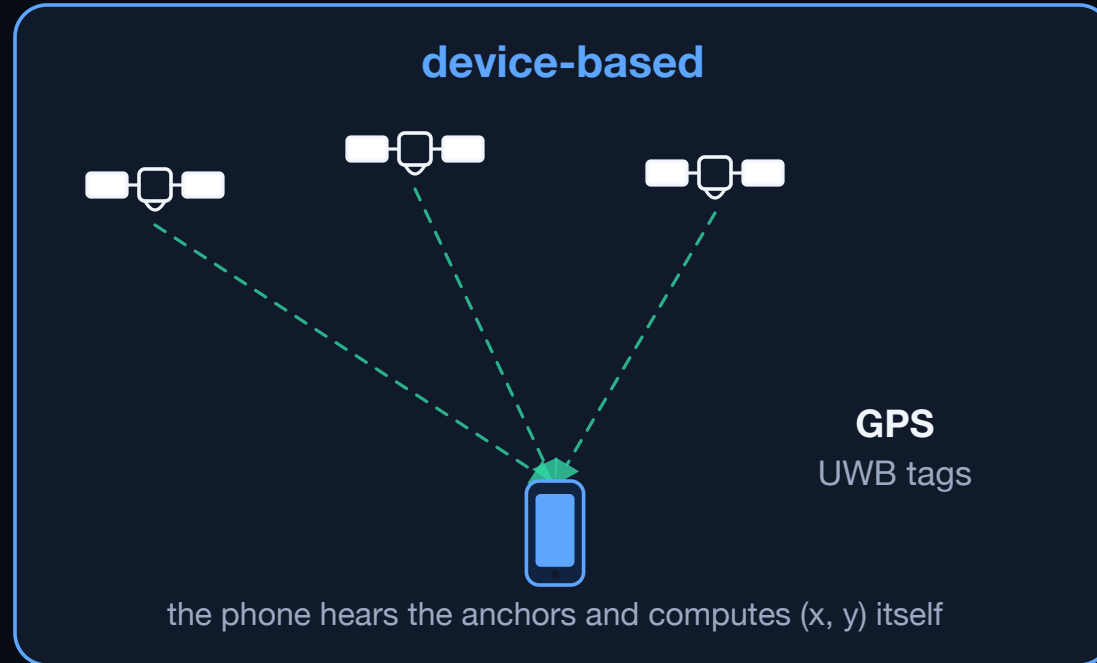
the phone's position (x, y)

Three anchors is the usual minimum in a plane.

Anchors: access points, beacons, satellites. Their positions are surveyed once and stored.

Known anchors, one measurement per signal, one unknown position.

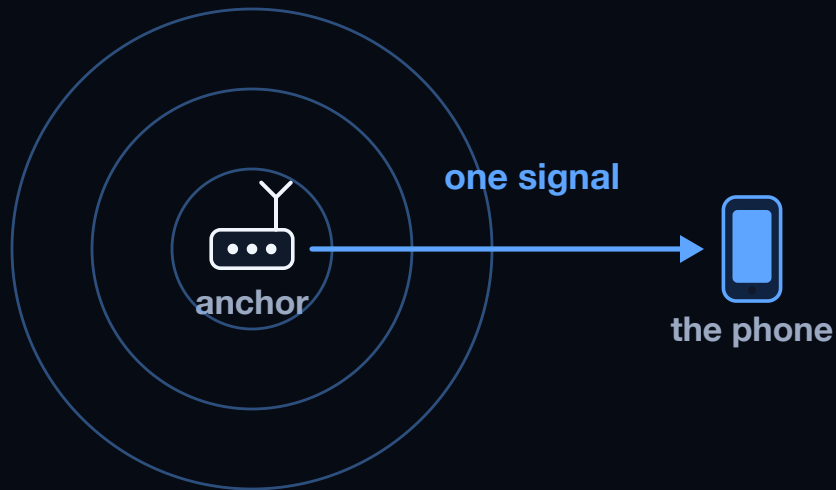
Who measures, and who computes?



The geometry is the same either way. The difference is which side holds the answer, and who has to carry the antennas.

Device-based: the phone knows where it is. Network-based: the network knows.

What in a received signal says where it came from?



how strong it is

received signal strength

→ a distance

on the floor plan: a circle



Part 1

when it arrived

time of flight

→ a distance

on the floor plan: a circle



Part 2

which way it came from

angle of arrival

→ a direction

on the floor plan: a line



Part 3

Strength, time, direction: each turns into a circle or a line on the plan.

How far away is the phone, from how loud it is?

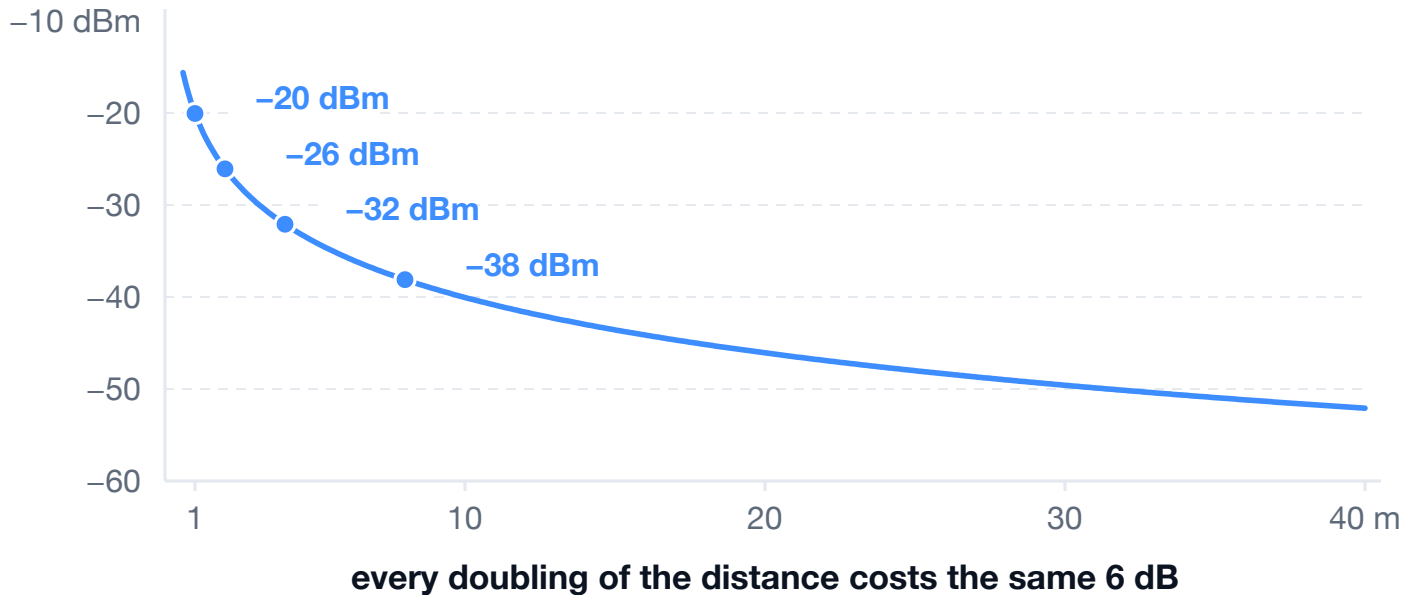
01

Distance from strength

Every receiver already reports how strong each packet arrived. This part turns that number into a distance, the distance into a circle, and three circles into a point, and then measures how far off that point is.

How does received power change with distance?

FREE SPACE, 2.4 GHz, +20 dBm OUT OF ONE ANTENNA INTO ANOTHER



$$P_r = P_t \cdot G_t \cdot G_r \cdot (\lambda / 4\pi d)^2$$

the Friis equation: power falls as $1 / d^2$

twice the distance -6 dB

ten times the distance -20 dB

the phone reports RSSI, in dBm

RSSI: received signal strength indicator.

Antenna gains $G_t = G_r = 1$ here. Indoors the exponent is steeper than 2, because walls absorb; the shape of the argument does not change.

Power falls as $1 / d^2$: read the power, and the distance follows.

1 Strength

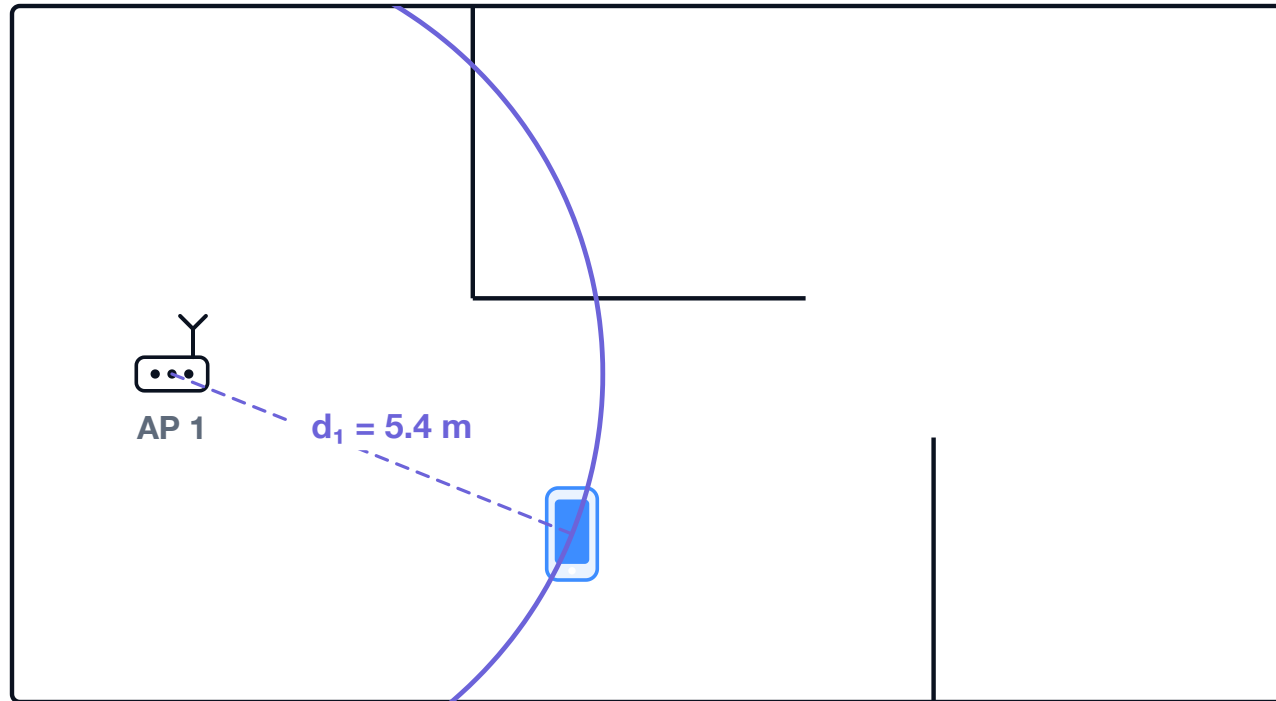
2 Time

3 Direction

4 Joint

5 Systems

One RSSI reading gives one circle



AP 1 hears the phone at -54.6 dBm

at 1 m it would be -40 dBm

so the distance is 5.4 m

$$d = 1 \text{ m} \cdot 10^{((P_0 - \text{RSSI}) / 20)}$$

One distance, and the phone is anywhere on the circle.

The AP does not know the direction.

One anchor: the phone is somewhere on a circle of radius d_1 .

1 Strength

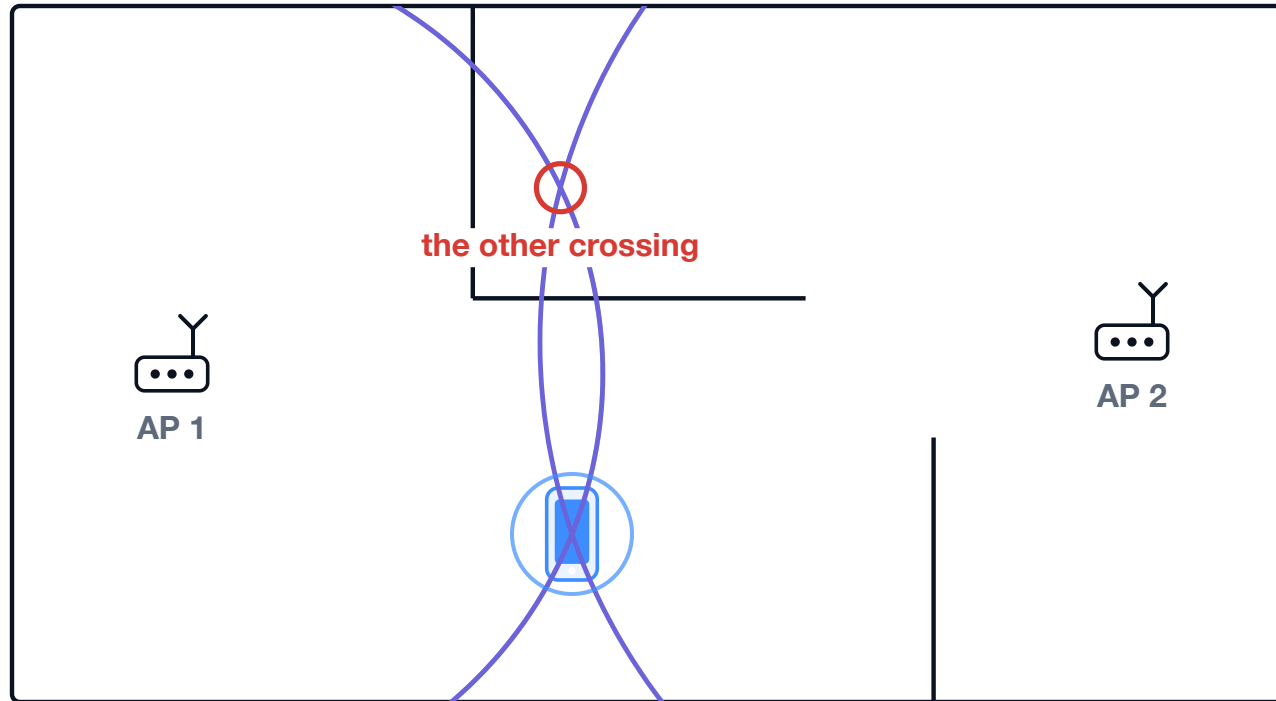
2 Time

3 Direction

4 Joint

5 Systems

Two circles cross at two points



AP 1 -54.6 dBm → 5.4 m

AP 2 -57.4 dBm → 7.4 m

Two circles cross twice.

One crossing is the phone. The other is a place the phone is not, and the two RSSI readings cannot tell them apart.

Two anchors leave two candidates.

1 Strength

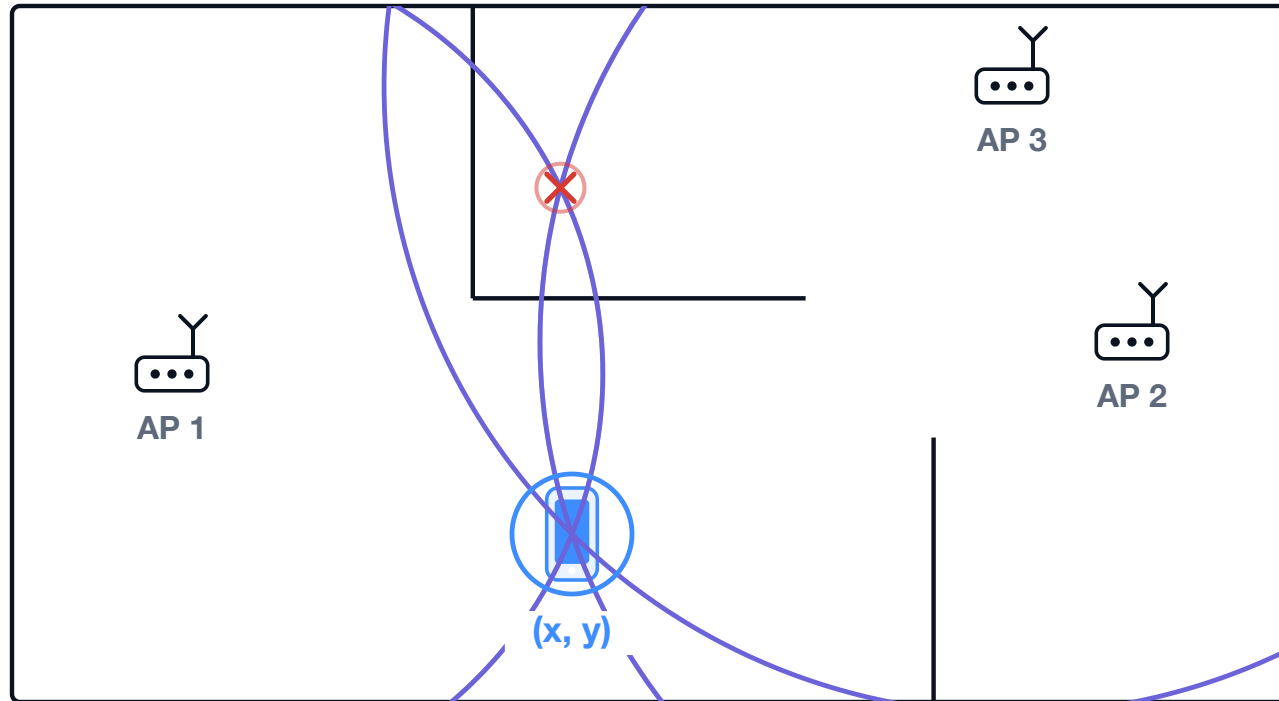
2 Time

3 Direction

4 Joint

5 Systems

Three circles cross at one point



AP 1 $-54.6 \text{ dBm} \rightarrow 5.4 \text{ m}$

AP 2 $-57.4 \text{ dBm} \rightarrow 7.4 \text{ m}$

AP 3 $-57.9 \text{ dBm} \rightarrow 7.8 \text{ m}$

trilateration

three distances to three known points fix one position

GPS does the same in three dimensions, with spheres.

Three anchors fix the position. This is trilateration.

1 Strength

2 Time

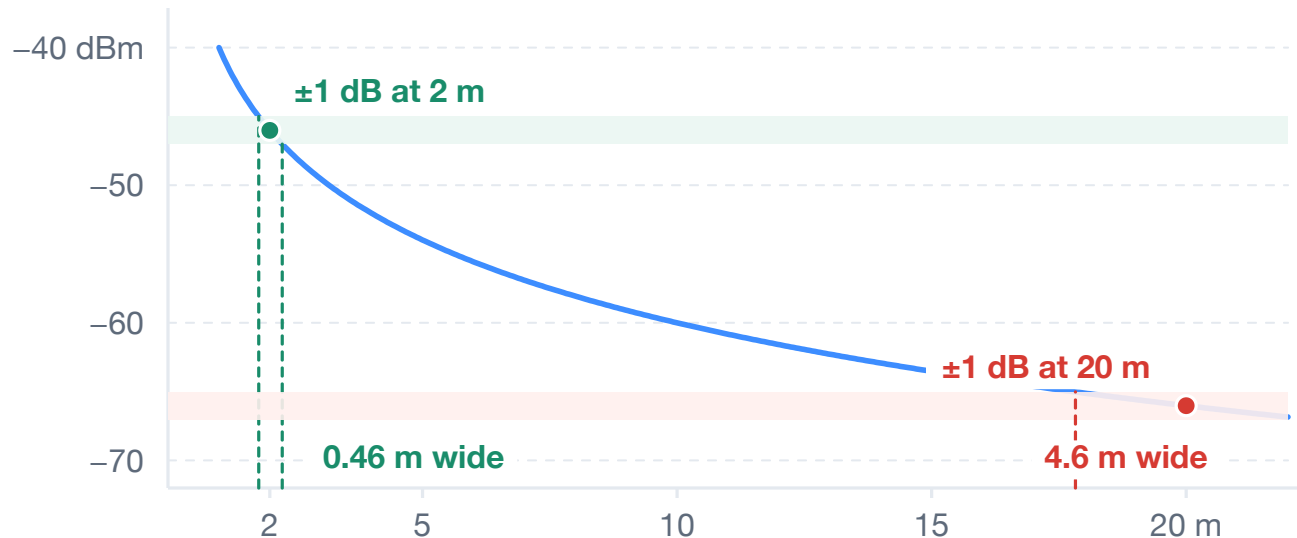
3 Direction

4 Joint

5 Systems

How much distance does 1 dB of error cost?

THE SAME ± 1 dB, READ AT 2 m AND AT 20 m



$$\Delta d / d = (\ln 10 / 20) \cdot \Delta P \approx 12 \% \text{ per dB}$$

1 dB at 2 m ± 0.24 m

1 dB at 20 m ± 2.4 m

3 dB at 20 m ± 8 m

The error is a fraction of the distance, so it grows with the distance. And indoor RSSI moves by far more than 1 dB, and the next four slides say why.

1 dB is 12 % of the distance: 24 cm at 2 m, 2.4 m at 20 m.

1 Strength

2 Time

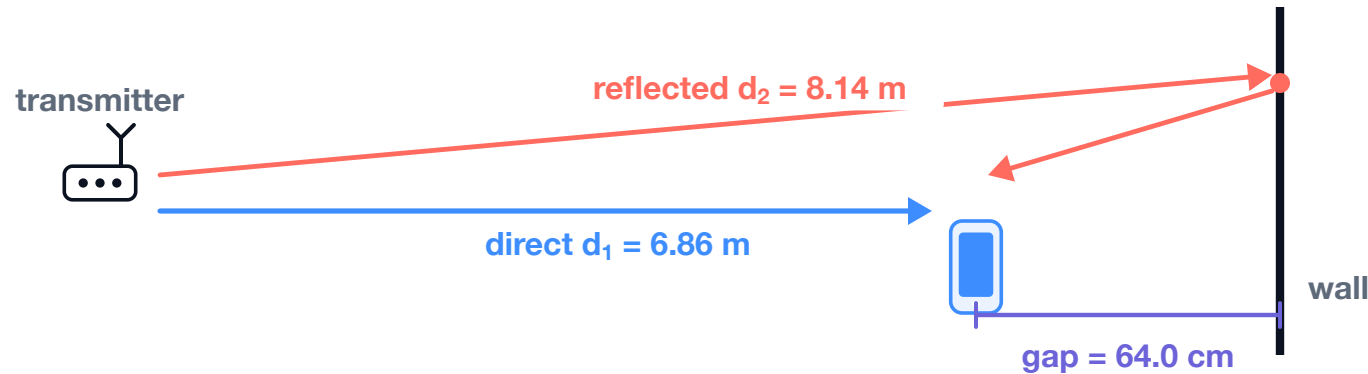
3 Direction

4 Joint

5 Systems

Where does a second copy come from?

ONE TRANSMITTER, ONE PHONE, AND A WALL A LITTLE WAY PAST IT



Walk 10 cm towards the wall: d_1 grows by 10 cm and d_2 shrinks by 10 cm.

one signal, sent once

the transmitter does not know it is there

two copies arrive

one straight, one off the wall and back

they differ only in how far

$\Delta d = 2 \times \text{gap} = 1.28 \text{ m}$, so 4.3 ns later

The two rays are drawn apart so both are visible; here they travel along the same line. A real room has one direct path and many reflections.

A reflection is the same signal, arriving later because it went farther.

1 Strength

2 Time

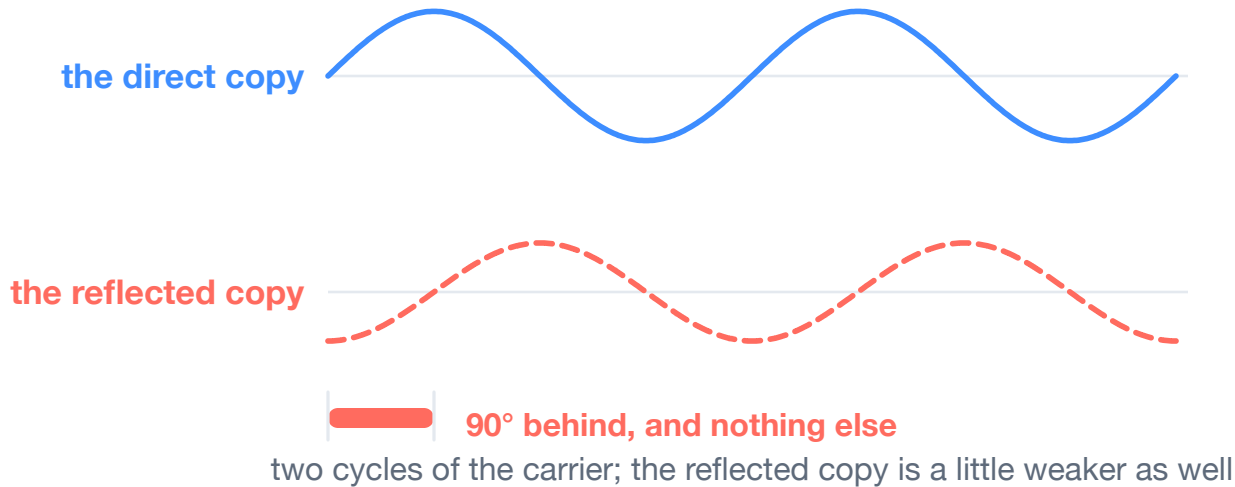
3 Direction

4 Joint

5 Systems

What does the extra path do to the copy?

AT 2.4 GHz ONE WAVELENGTH IS 12.5 cm AND ONE CYCLE LASTS 0.42 ns



$$\Delta\phi = 2\pi \cdot \Delta d / \lambda$$

one full turn of phase per wavelength of extra path

the extra path

$\Delta d = 1.280 \text{ m}$

in wavelengths

10.25λ

whole turns look the same

10 of them

what is left

90° behind

Only the leftover fraction of a turn can be seen.

The extra path turns the copy by $2\pi \Delta d / \lambda$: one full turn per wavelength.

1 Strength

2 Time

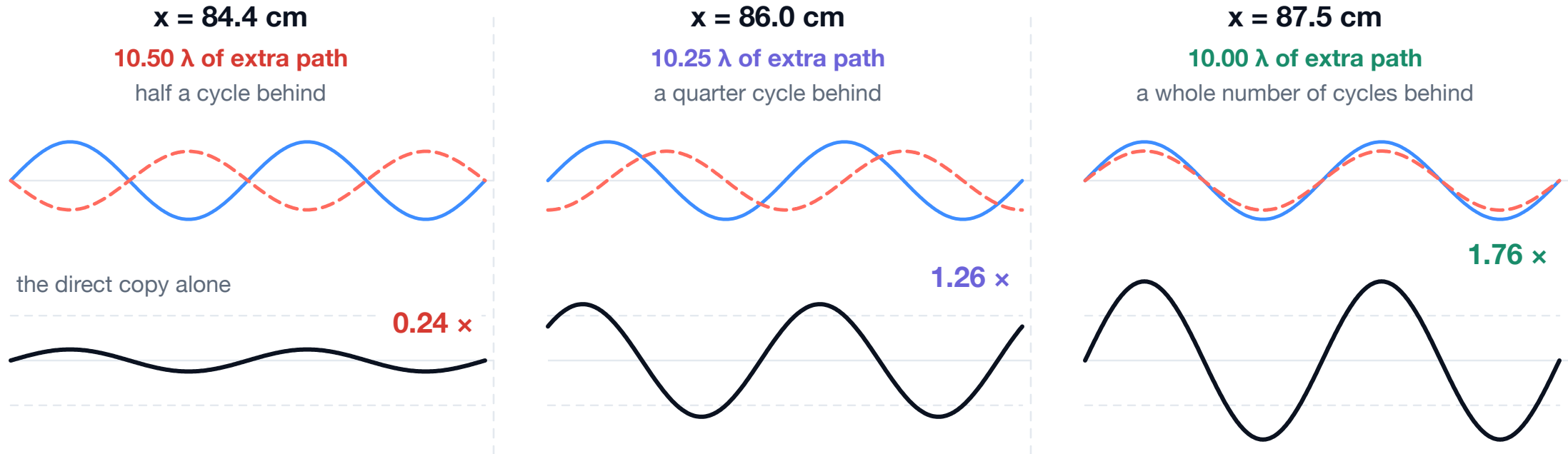
3 Direction

4 Joint

5 Systems

Two copies add, and the phase difference decides the sum

THE SAME TWO PATHS AT THREE POSITIONS, THE PHONE WALKING 1.6 cm EACH TIME



Top row: the two copies that arrive, the direct one blue and the reflected one dashed. Bottom row: their sum, against the dashed line that marks the direct copy on its own.

Three centimetres of walking, and -12 dB becomes +5 dB.

1 Strength

2 Time

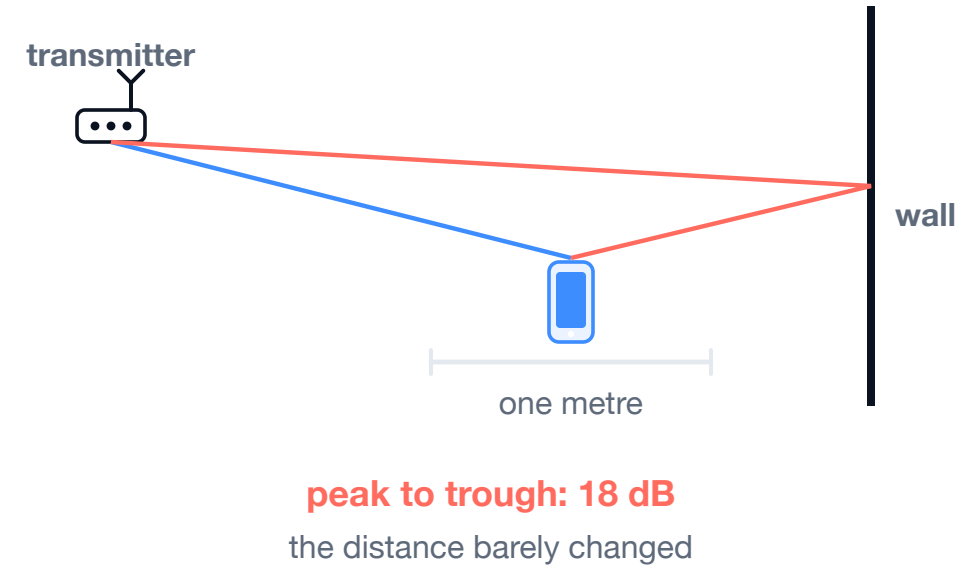
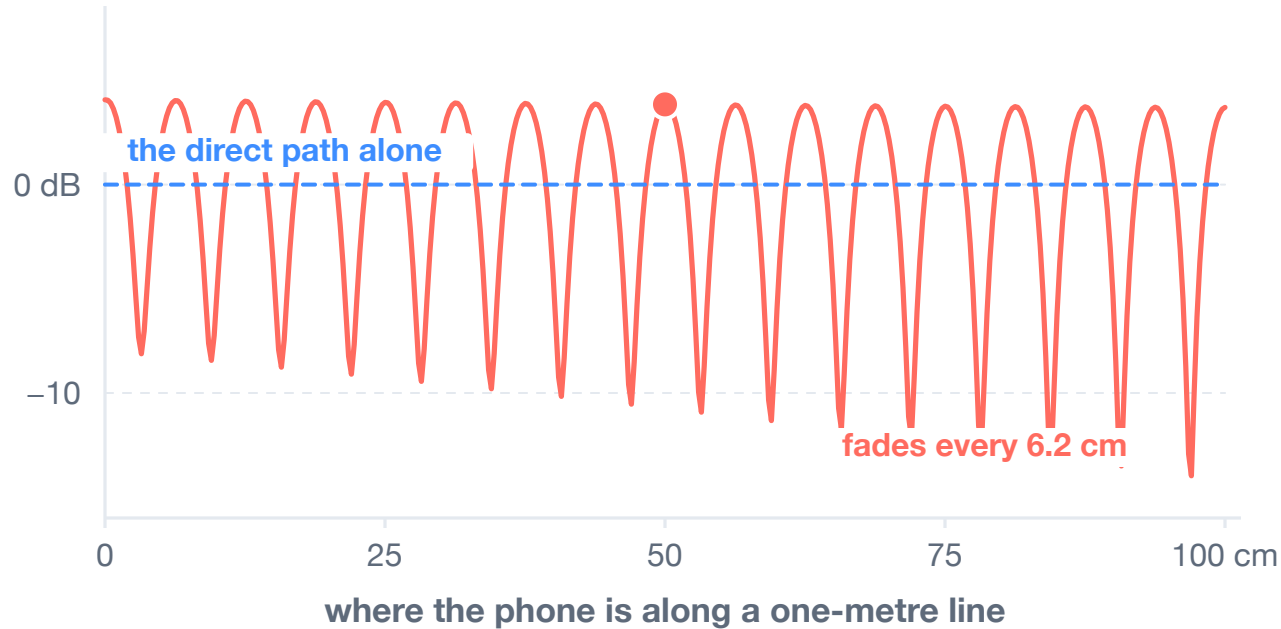
3 Direction

4 Joint

5 Systems

What does that look like over a whole metre?

THE SAME TWO PATHS, SWEEPED ALONG THE WHOLE ONE-METRE LINE



One metre of walking, two paths, and the signal swings by 18 dB.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

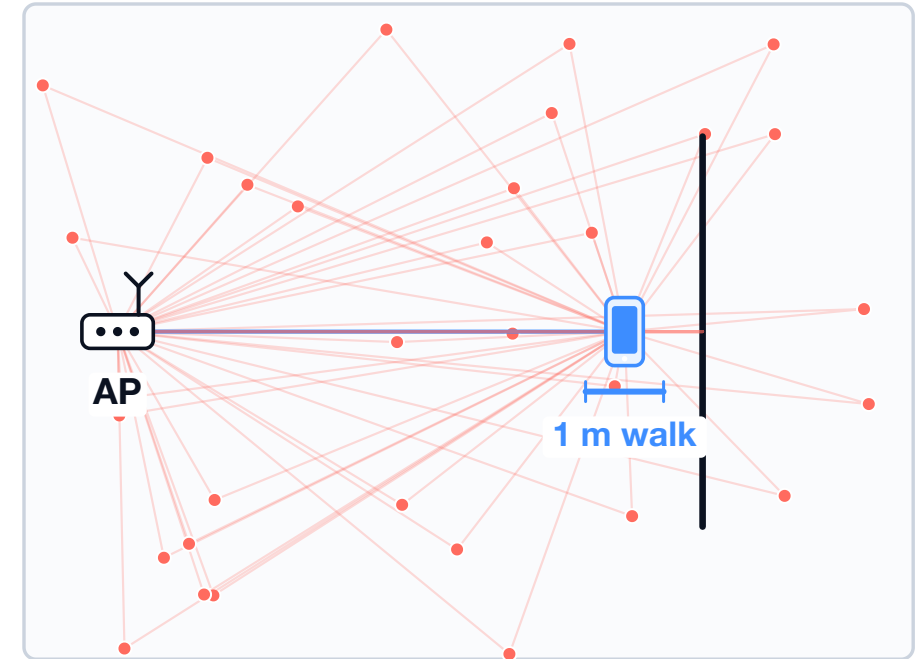
What does a real room do to that curve?

THE SAME ONE-METRE WALK, NOW WITH MANY ARRIVALS FROM DIFFERENT DIRECTIONS



Irregular peak spacing and fade depth

Illustrative channel: 34 coherent paths with unequal strengths and fixed reflection phases. As the phone moves, their different path-length changes create irregular fading.



With many paths, RSSI has no simple repeating pattern along the walk.

1 Strength

2 Time

3 Direction

4 Joint

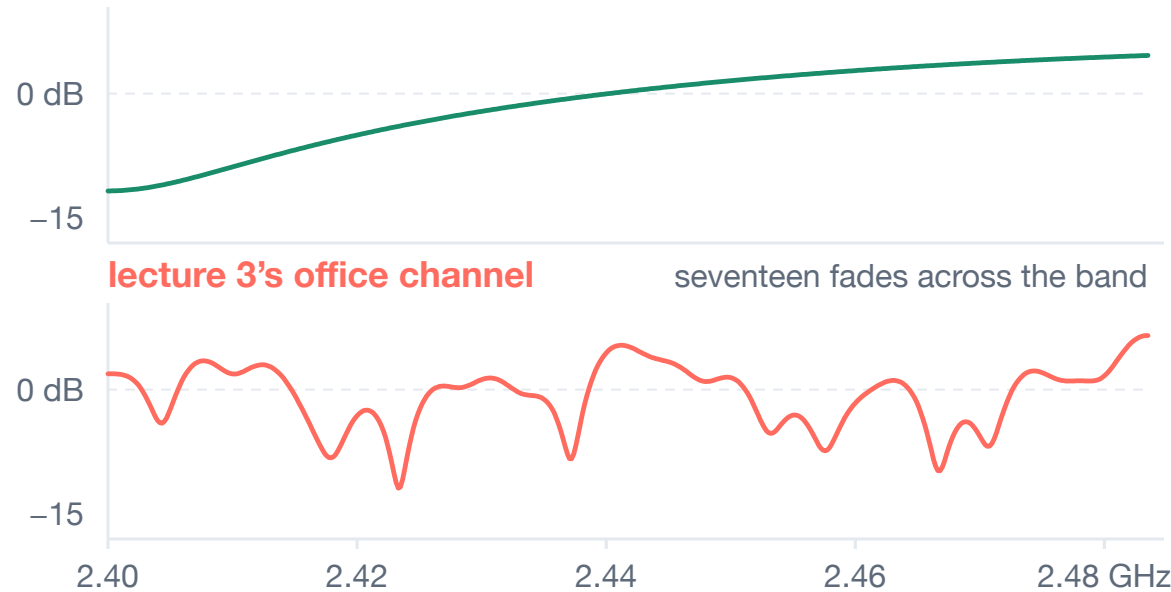
5 Systems

Or hold the phone still and change the frequency

THE PHONE HELD STILL AT $x = 84.4$ cm, AND THE CARRIER SWEEPED ACROSS THE WHOLE 2.4 GHz BAND

one reflection, 1.31 m longer

a third of a fade across the band



one 20 MHz Wi-Fi channel

$$\Delta\phi = 2\pi \cdot \Delta d \cdot f / c$$

the same Δd ; now it is λ that moves

$\Delta d = 1.31$ m

229 MHz per fade

$\Delta d = 63$ m

5 MHz per fade

one Wi-Fi channel

20 MHz wide

This is the fading of lecture 3

and it is why one wide channel had to become many narrow subcarriers.

Δd is fixed by the room; $\Delta\phi$ is Δd measured in wavelengths. Moving the phone changes Δd , changing the carrier changes λ , and both turn the same arrow.

Position or frequency: the same two paths fade either way.

1 Strength

2 Time

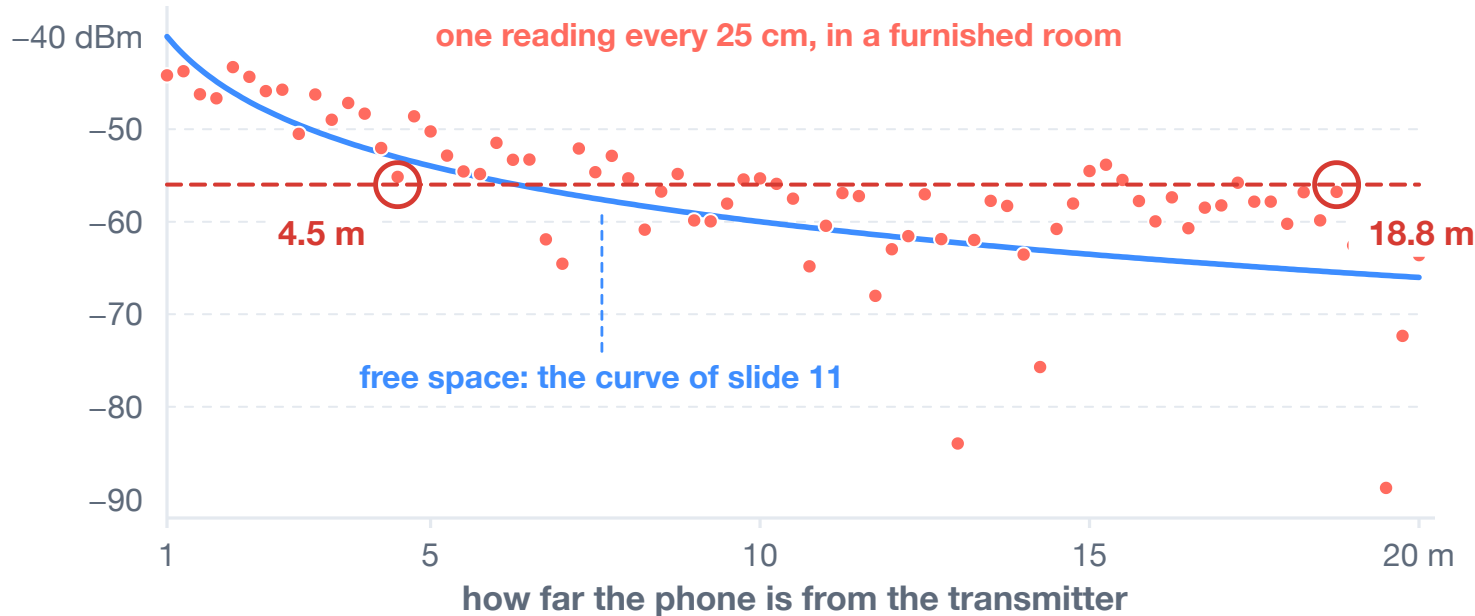
3 Direction

4 Joint

5 Systems

Does the $1 / d^2$ curve survive a real room?

ONE TRANSMITTER, ONE PHONE CARRIED FROM 1 m TO 20 m, ONE READING EVERY 25 cm



what the room adds -23 to +10 dB

-56 dBm is read at 11 of the 77 spots

those spots run from 4.5 m to 18.8 m

the inverse does not exist

Slide 12 turned one RSSI reading into one distance. In this room that step has eleven answers.

In a real room the strength no longer tells you the distance.

1 Strength

2 Time

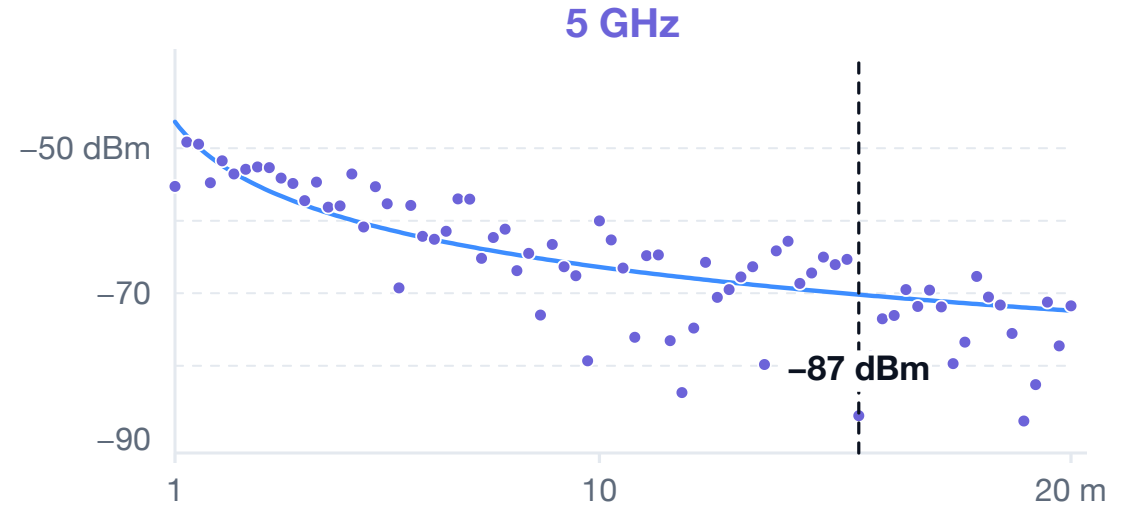
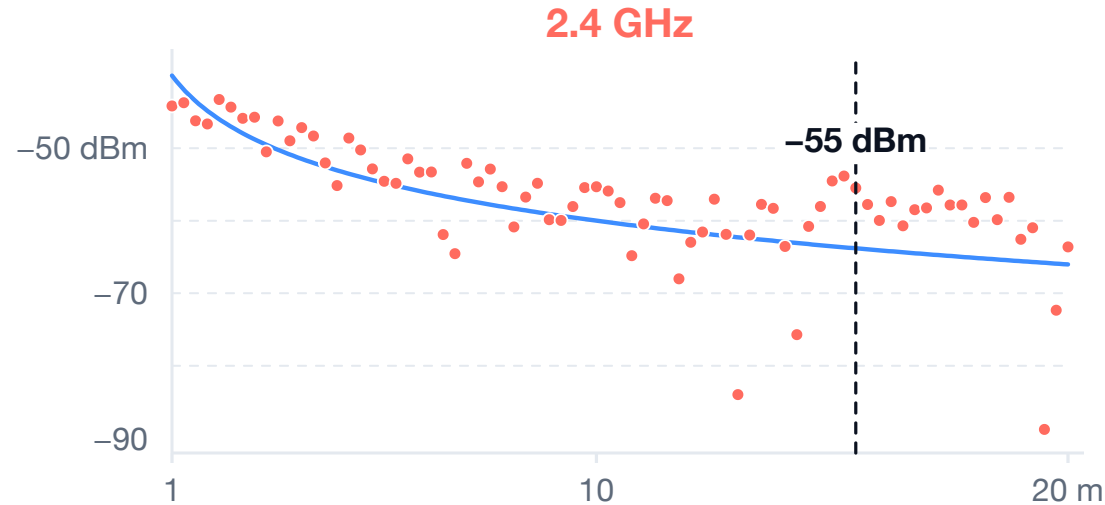
3 Direction

4 Joint

5 Systems

The same walk, at two frequencies

THE SAME ROOM, THE SAME WALK, THE SAME INSTANT — TWO CARRIERS



at 15.5 m the two bands disagree by 31 dB

theory: the same shape, 6.4 dB apart

measured: -18 to +31 dB apart, spot by spot

Free space says the 5 GHz curve is the 2.4 GHz curve moved down by $20 \log_{10}(5 / 2.4) = 6.4$ dB, and nothing else. The wavelength is different, so every path lands on a different phase.

Change the band and the same room gives a different map.

1 Strength

2 Time

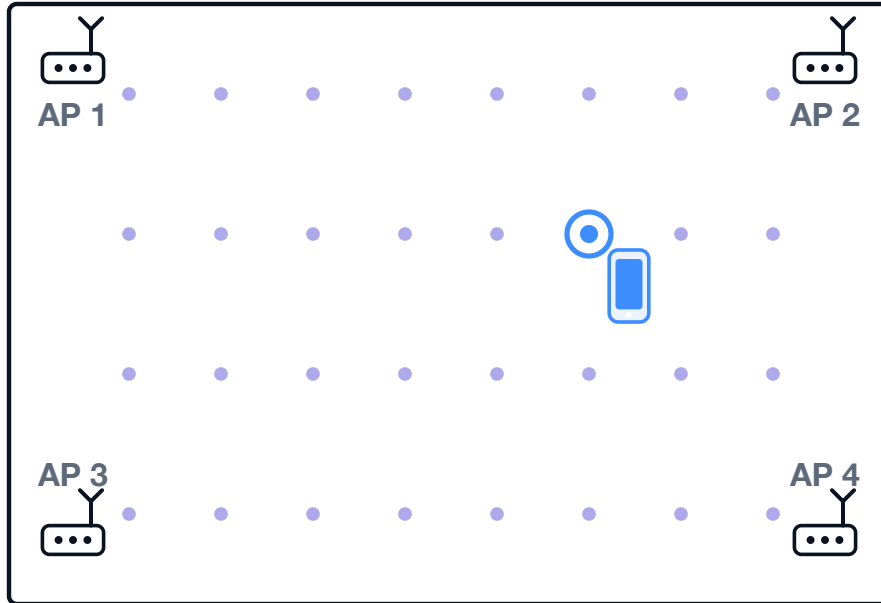
3 Direction

4 Joint

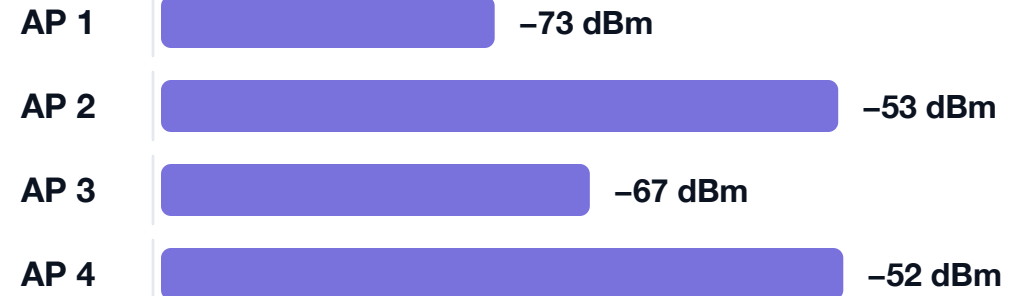
5 Systems

What if we stop converting RSSI to distance?

STOP ASKING WHAT DISTANCE AN RSSI MEANS — JUST WRITE THE FOUR NUMBERS DOWN



the fingerprint of this one spot



no distance is computed anywhere — the four numbers are the record

the survey: walk the grid, stand still, record, move on

32 points here; a real building has thousands

Fingerprinting: record the four numbers at every spot, before anyone asks.

1 Strength

2 Time

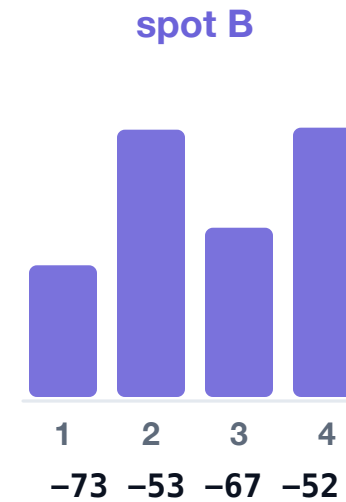
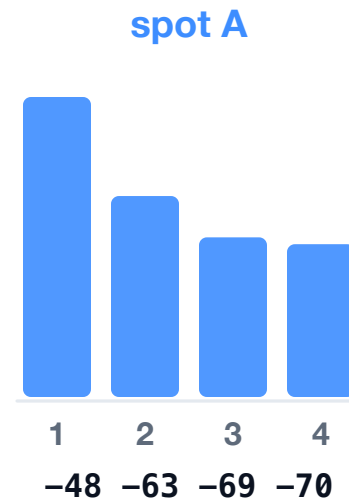
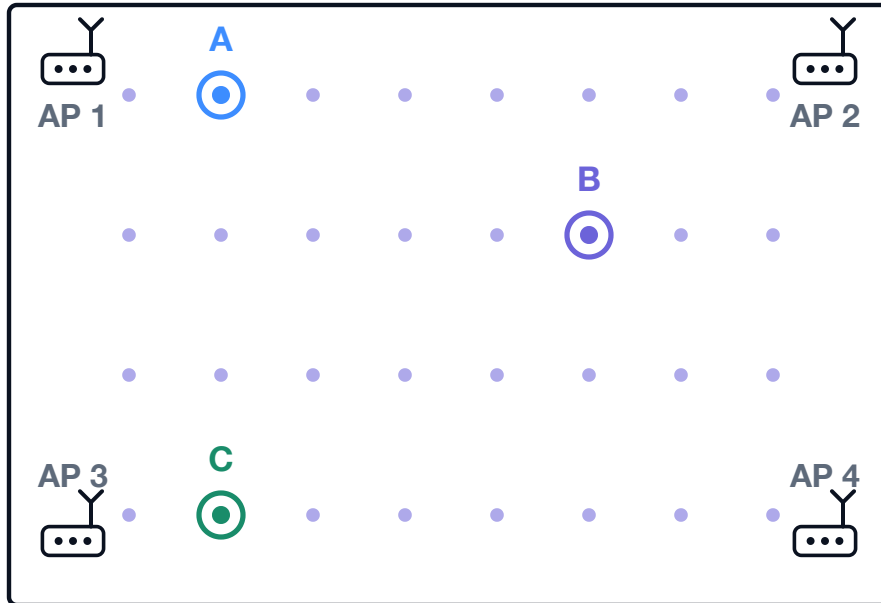
3 Direction

4 Joint

5 Systems

Why is that a fingerprint?

THE SAME FOUR APs, THREE SPOTS IN THE SAME ROOM



one AP alone only says “somewhere on a circle”; the four of them together say “here”.

The fades are different at every point in the room, and they stay where they are for as long as the room does. That is what makes the pattern a name for the place — and why the survey has to be redone when the furniture moves.

The multipath that ruined RSSI as a distance is what makes it a good name for a spot.

1 Strength

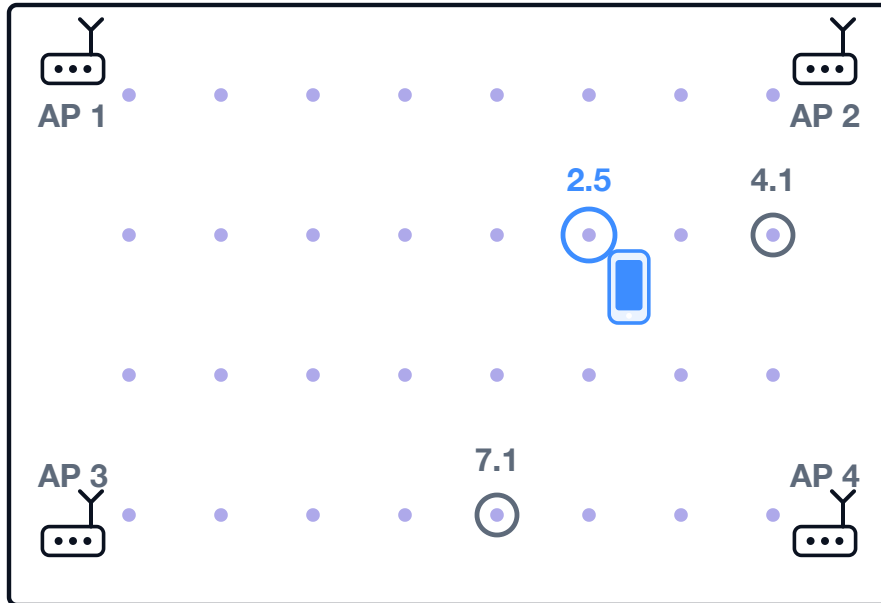
2 Time

3 Direction

4 Joint

5 Systems

How does the phone find itself in the map?



what the phone hears now

[-71, -54, -66, -54] dBm

distance to each stored vector, in dB

(7.2 m, 2.9 m) 2.5 dB

(9.6 m, 2.9 m) 4.1 dB

(6.1 m, 6.4 m) 7.1 dB

position = the grid point whose fingerprint is nearest

Nearest neighbour in RSSI space. Averaging the k nearest points softens the grid; a model of the RSSI distribution does better still.

No distance is ever computed. The pattern is matched, not inverted.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

What does the map cost?

WHAT FINGERPRINTING BUYS, AND WHAT IT COSTS

it works with multipath

the fades are part of the fingerprint, not an error

the AP positions are never needed

the survey

every grid point, every floor, once per building

hours to days of walking with a phone

the upkeep

a moved cabinet, a new AP, a crowd: the vectors change

the map ages, and has to be walked again

RADAR, 2000

the first Wi-Fi fingerprinting system

2 to 3 m

median error, three access points

Twenty-five years later, phones still do this
for the coarse fix, before anything finer.

Bahl and Padmanabhan, RADAR: an in-building RF-based user location and tracking system, INFOCOM 2000. The 2–3 m figure is the median they report.

Fingerprinting trades multipath for a survey that has to be kept current.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

What strength cannot give us

WHERE PART 1 ENDS

method	typical error	what limits it
RSSI → distance → circles	it does not work	one reading fits eleven distances, 4.5 m to 18.8 m
RSSI fingerprinting	2 – 3 m	the survey, the map going stale, one map per band

For a metre we need a quantity that does not fade.

Two candidates from the map on slide 9: when the signal arrived, and which way it came from.

time of flight

Part 2: 1 ns is 30 cm

angle of arrival

Part 3: the phase steps across an array, read backwards

Strength gives the room. For the metre, measure time or direction.

1 **Strength**

2 Time

3 Direction

4 Joint

5 Systems

How far away is the phone, from when the signal arrived?

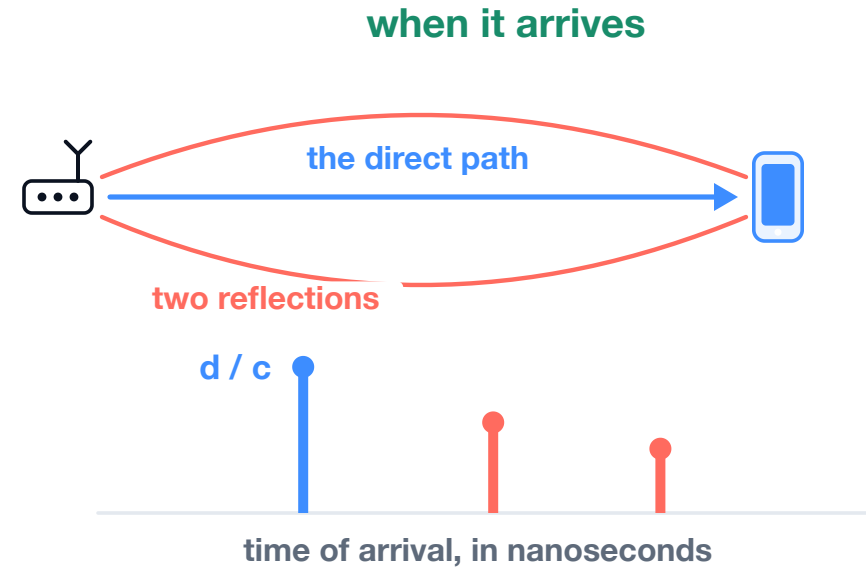
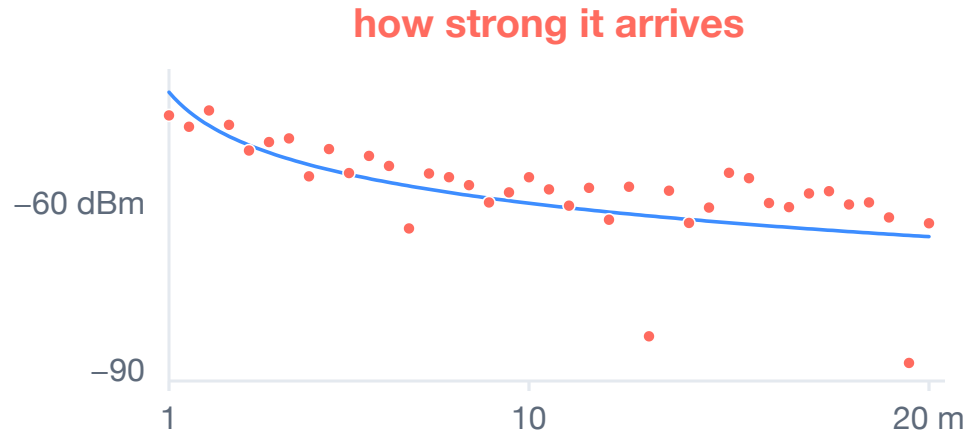
02

Distance from time

Radio covers 30 cm in a nanosecond, and that number does not fade. This part asks what it takes to time an arrival to a nanosecond: whose clock, how many trips, and how much bandwidth.

Why should time do better than strength?

THE SAME ROOM, TWO QUANTITIES TO MEASURE IT WITH



✗ one reading fits eleven distances, and the band changes it ✓ the first arrival is the direct path, and it is exactly d / c

A reflection can add to the signal or cancel it, so the strength has no fixed meaning. A reflection can only arrive later than the direct path, never before it, so the first arrival still means the distance.

Strength has no fixed meaning. The first arrival always means d / c .

1 Strength

2 Time

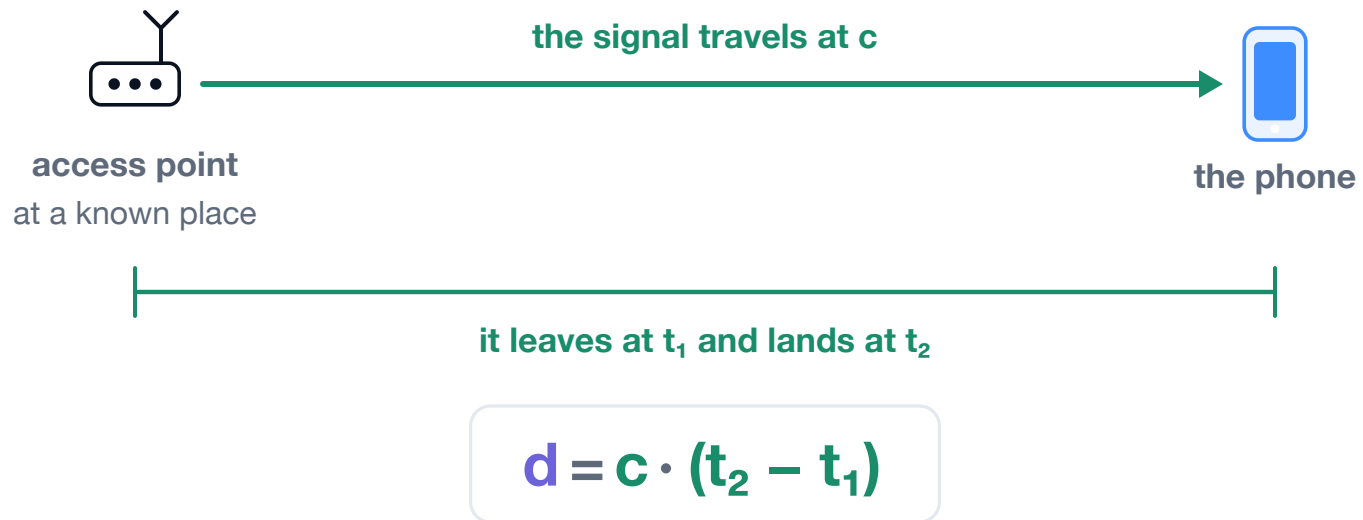
3 Direction

4 Joint

5 Systems

How do you turn a time into a distance?

THE WHOLE METHOD, IN ONE LINE



c 30 cm per ns

$t_2 - t_1 = 26.7$ ns $d = 8.0$ m

$t_2 - t_1 = 66.7$ ns $d = 20.0$ m

nothing in the room
changes c , and nothing
changes the length of the
straight line.

This is what GPS does with satellites, and what a tape measure would do if light were slow enough to watch.

Distance is the travel time times the speed of light.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

How far does radio travel in one nanosecond?

$c = 299,792,458 \text{ m/s}$, ROUNDED TO 30 cm PER NANOSECOND

1 ns 30 cm a shoe	10 ns 3 m a room	100 ns 30 m a building	1 μs 300 m a city block	1 ms 300 km a state
---------------------------------------	--------------------------------------	--	---	---

To place the phone to 30 cm, the arrival time has to be known to 1 ns.

A clock that is 1 μs wrong puts the phone 300 m away, in another building.

Thirty centimetres of accuracy needs the arrival time to one nanosecond.

1 Strength

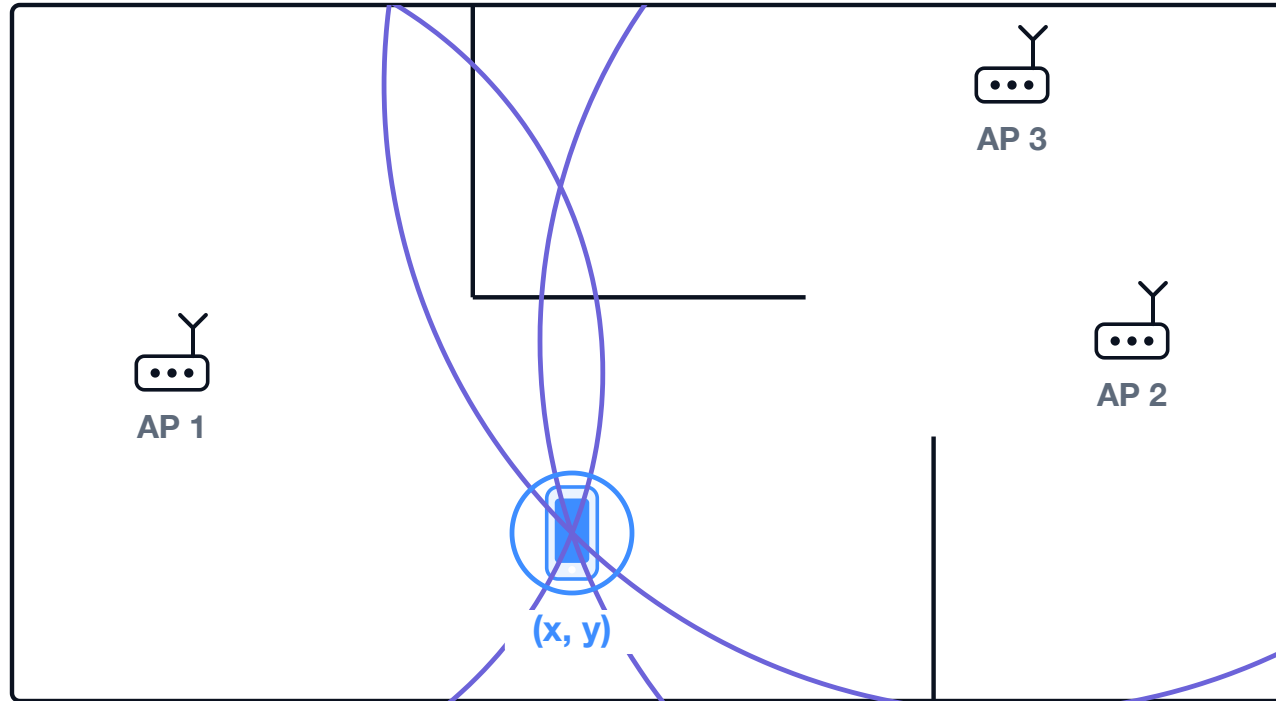
2 Time

3 Direction

4 Joint

5 Systems

Three flight times, three circles



AP 1 18.0 ns → 5.4 m

AP 2 24.7 ns → 7.4 m

AP 3 26.2 ns → 7.8 m

the same picture as slide 14

three distances, three circles, one crossing

and how does the phone get $t_2 - t_1$?

The anchors are the same anchors, surveyed once, and the crossing is found the same way. Only the ruler has changed: a time instead of a strength.

The geometry of Part 1 is reused untouched. Only the ruler is new.

1 Strength

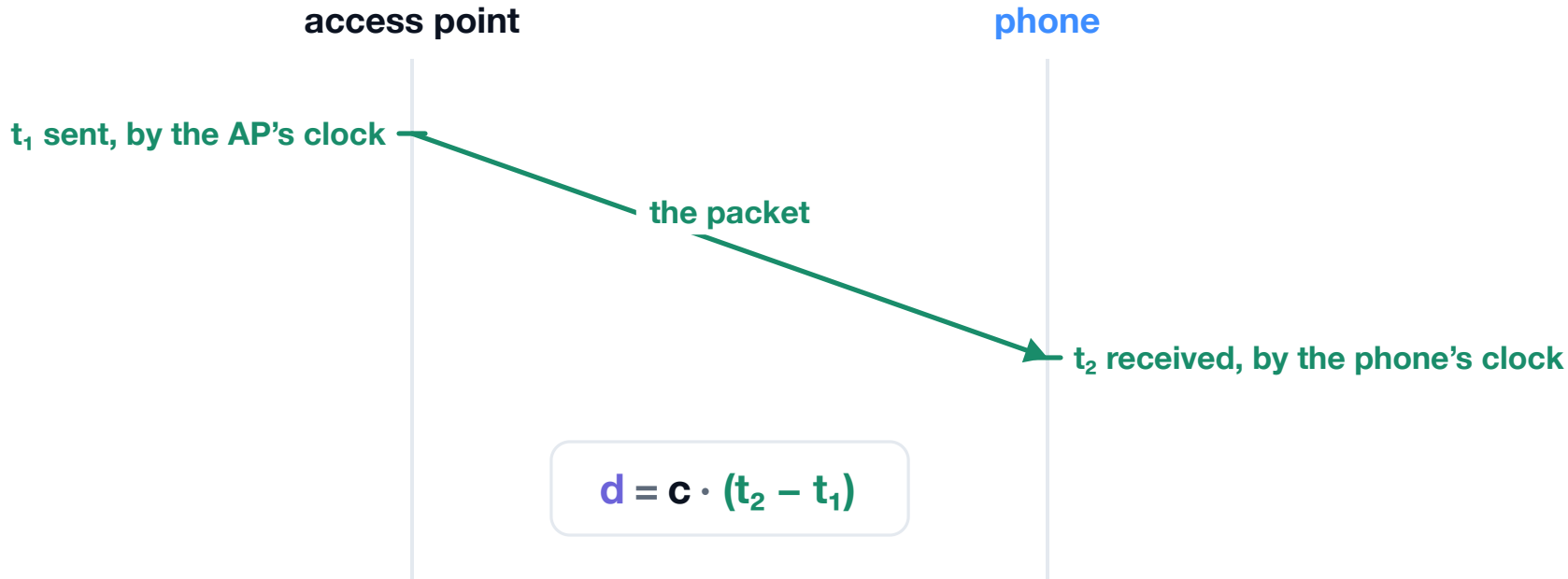
2 Time

3 Direction

4 Joint

5 Systems

Can the phone just read the arrival time?



two clocks

t_1 is read on one clock,
 t_2 on another.

The difference contains the
flight time, plus whatever
the two clocks disagree by.

One-way time of flight needs both clocks to agree to 1 ns.

1 Strength

2 Time

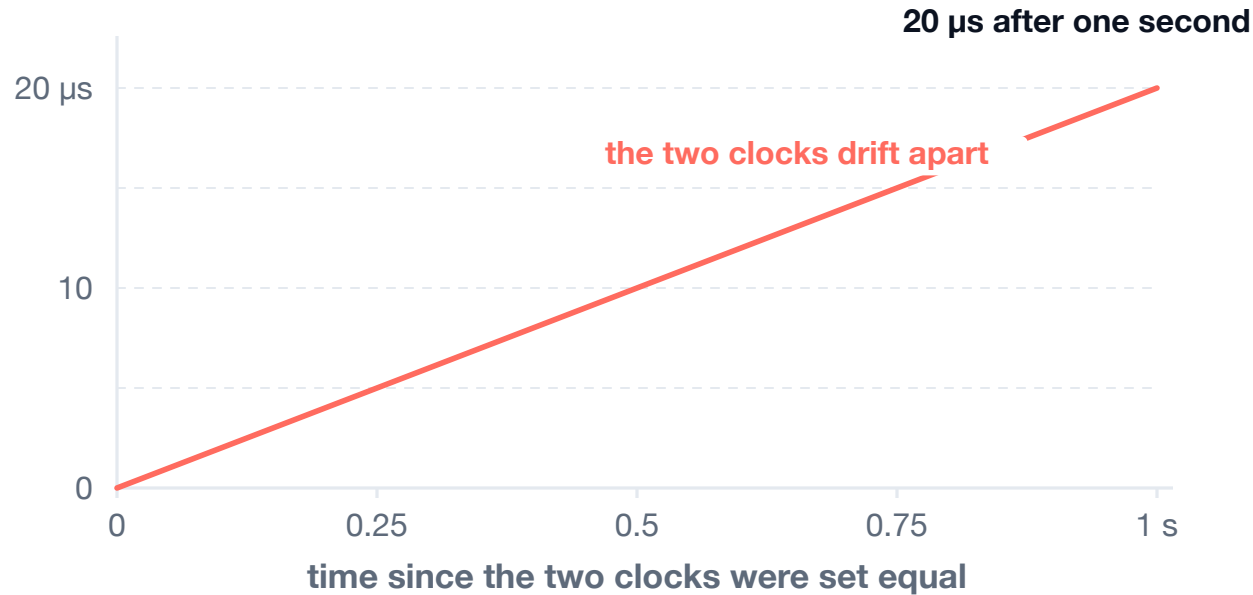
3 Direction

4 Joint

5 Systems

How far apart are two clocks?

AN ORDINARY CRYSTAL OSCILLATOR IS RATED AT ABOUT 20 PARTS PER MILLION



20 ppm $\times$ 1 s

20 μ s = 6 km

the target

1 ns = 30 cm

so the clocks must agree to 1 part in 10^{12}

How GPS gets away with it

atomic clocks in the satellites, and the receiver's clock offset solved as a fourth unknown: 4 satellites

A phone and an access point have no atomic clock and no fourth satellite. One-way timing between them is out.

Ordinary clocks drift kilometres per second. One-way timing needs a trick.

1 Strength

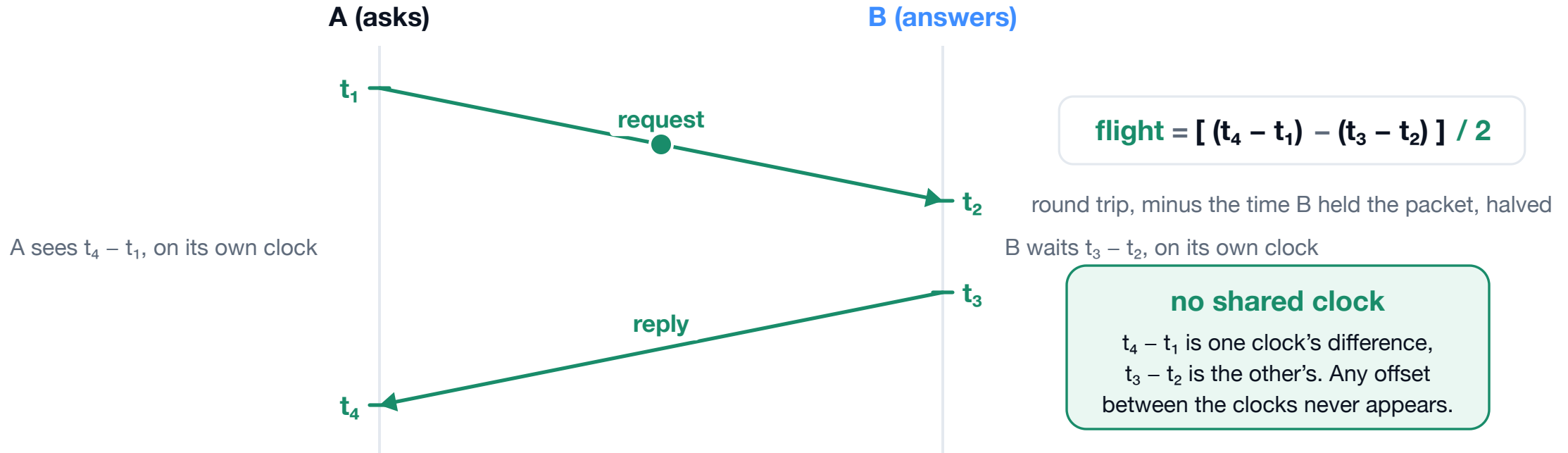
2 Time

3 Direction

4 Joint

5 Systems

What if the signal goes there and back?



Two-way ranging. What is left is the drift of one clock over one round trip: 20 ppm of a few hundred microseconds is a few nanoseconds.

Each side subtracts only its own timestamps. The clock offset cancels.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

Two-way ranging on real radios

THE SAME FOUR TIMESTAMPS, IN TWO STANDARDS

Wi-Fi FTM

802.11mc “fine timing measurement”, 2016

- built into ordinary access points and phones
- exposed to apps on Android as “Wi-Fi RTT”
- channel bandwidth 20 to 160 MHz
- typical error: one to two metres

UWB

802.15.4z, ultra-wideband ranging

- a separate radio: phones since 2019, tags since 2021
- channels 500 MHz wide, around 6 to 8 GHz
- the same request–reply exchange, called DS-TWR
- typical error: about ten centimetres

Same protocol. A factor of ten in accuracy. The next slide says where the factor comes from.

Both Wi-Fi and UWB range this way. UWB gets ten times closer.

1 Strength

2 Time

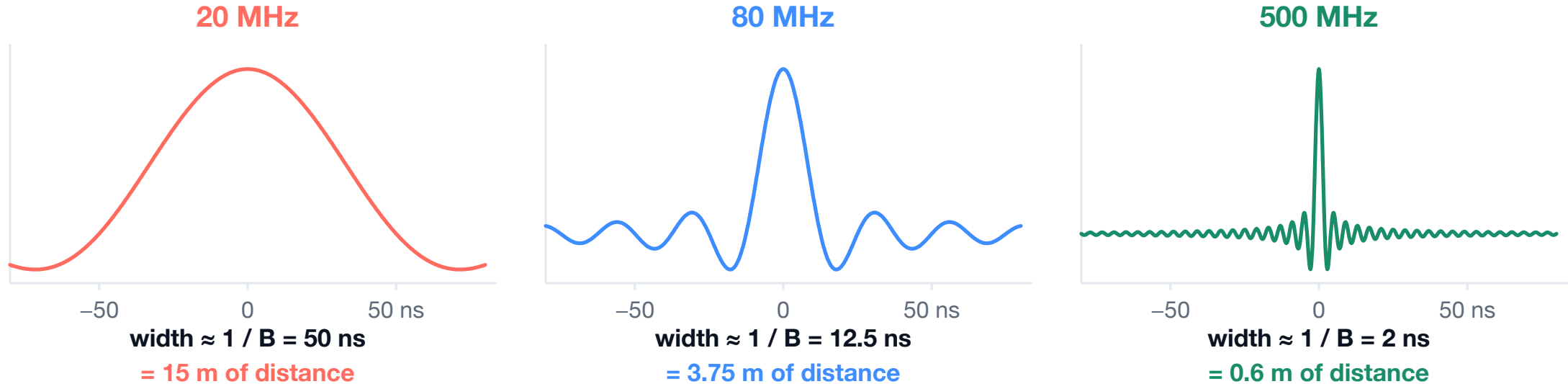
3 Direction

4 Joint

5 Systems

How sharply can a receiver see an arrival?

THE SHORTEST PULSE A CHANNEL OF BANDWIDTH B CAN CARRY IS ABOUT $1 / B$ LONG



The receiver timestamps the peak. A broad peak is a vague timestamp.

Timing resolution $\approx 1 / B$: 50 ns at 20 MHz, 2 ns at 500 MHz.

1 Strength

2 Time

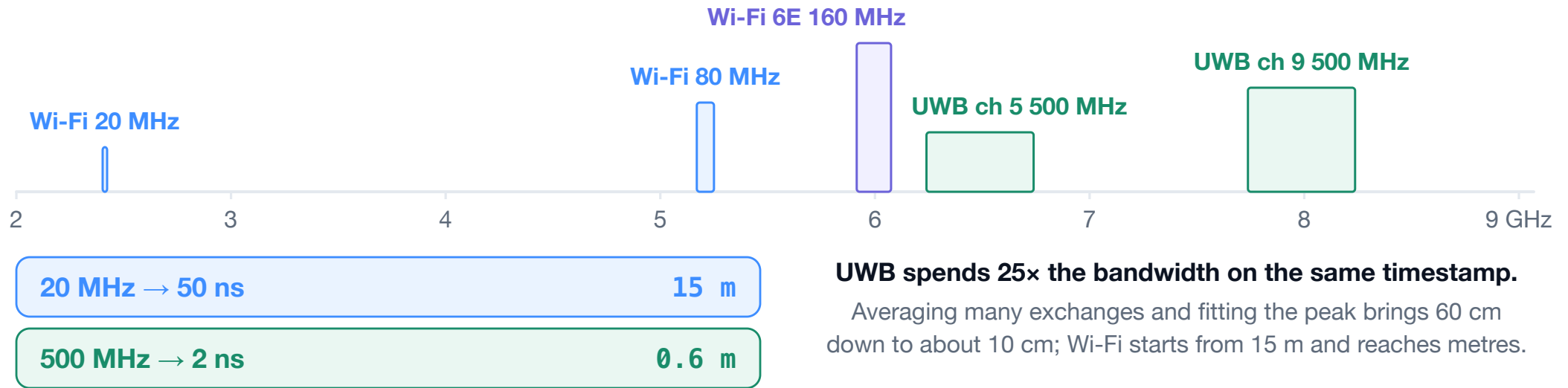
3 Direction

4 Joint

5 Systems

Why UWB is 10 cm and Wi-Fi is metres

CHANNEL WIDTHS DRAWN TO SCALE ON ONE FREQUENCY AXIS



Bandwidth buys time resolution. UWB spends 500 MHz on it.

1 Strength

2 Time

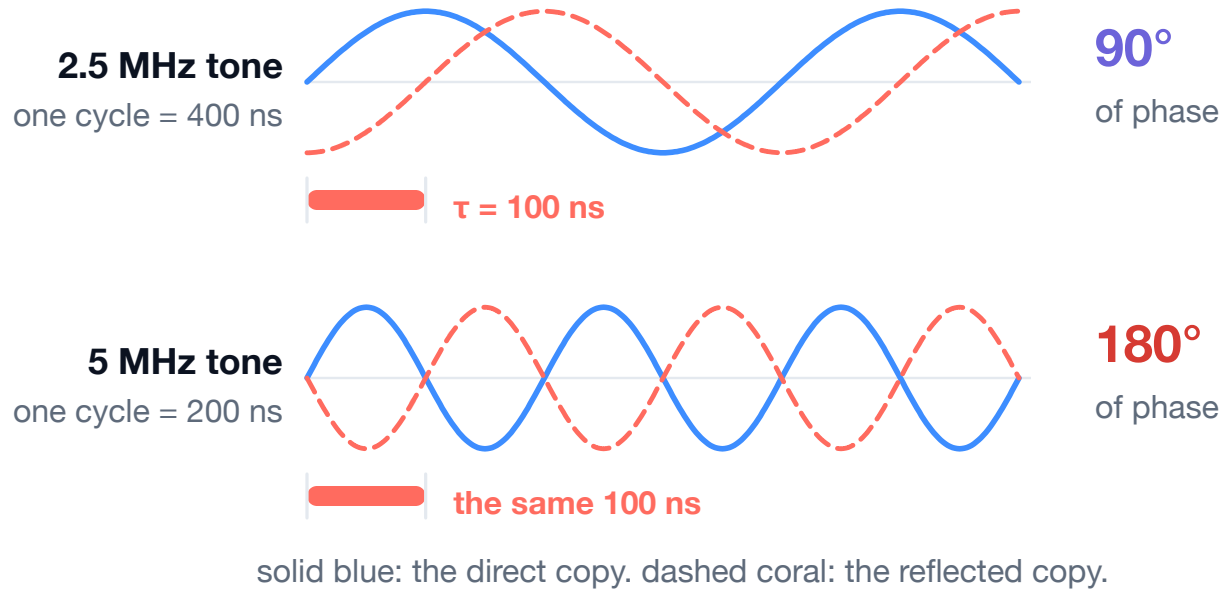
3 Direction

4 Joint

5 Systems

OFDM sends tones, not pulses — where is the delay?

ONE REFLECTION, $\tau = 100$ ns LATER — THE SAME DELAY ON TWO DIFFERENT TONES



$$\angle H_k = -2\pi \cdot f_k \cdot \tau$$

the phase a delay of τ costs at frequency f_k

the delay

$\tau = 100$ ns

at 2.5 MHz

90° of phase

at 5.0 MHz

180° of phase

and the phone already measures it

$\angle H_k$ is in every channel estimate it makes

Half a cycle of delay turns the copy upside down, which is the fade of Part 1 again. The two tones are drawn far apart so the difference is visible; real neighbouring subcarriers are only 312.5 kHz apart.

The same delay is a bigger phase at a higher frequency.

1 Strength

2 Time

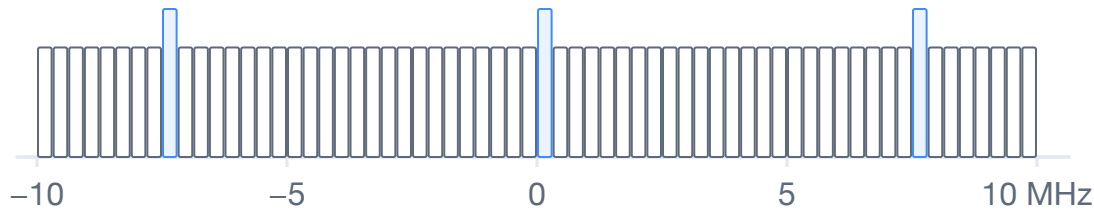
3 Direction

4 Joint

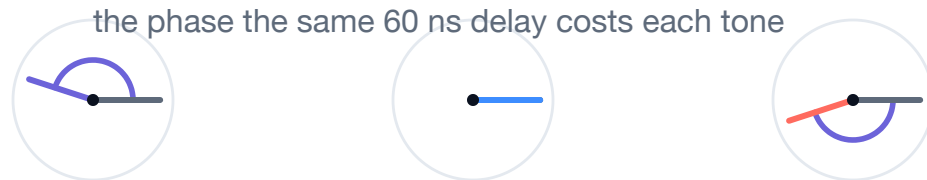
5 Systems

Why does one delay give every subcarrier a different phase?

LECTURE 3: ONE 20 MHz CHANNEL IS 64 SUBCARRIERS, EACH A TONE OF ITS OWN



64 tones, 312.5 kHz apart, all sent in the same symbol



subcarrier -24
+162°

subcarrier 0
+0°

subcarrier +24
-162°

the phase the same 60 ns delay costs each tone

$$\angle H_k = -2\pi \cdot f_k \cdot \tau$$

slide 39, unchanged: bigger f_k , bigger phase

every tone has its own f_k so its own phase

neighbours differ by $\Delta f = 312.5$ kHz

so the phase step is 6.8° per subcarrier

slide 39 used two tones 2.5 MHz apart
these 64 are the same law in tiny even steps

The dials are drawn against the middle tone, because only differences between tones can be seen: the part they all share is absorbed by the radio's own unknown offset.

Each subcarrier is a different frequency, so one delay costs each a different phase.

1 Strength

2 Time

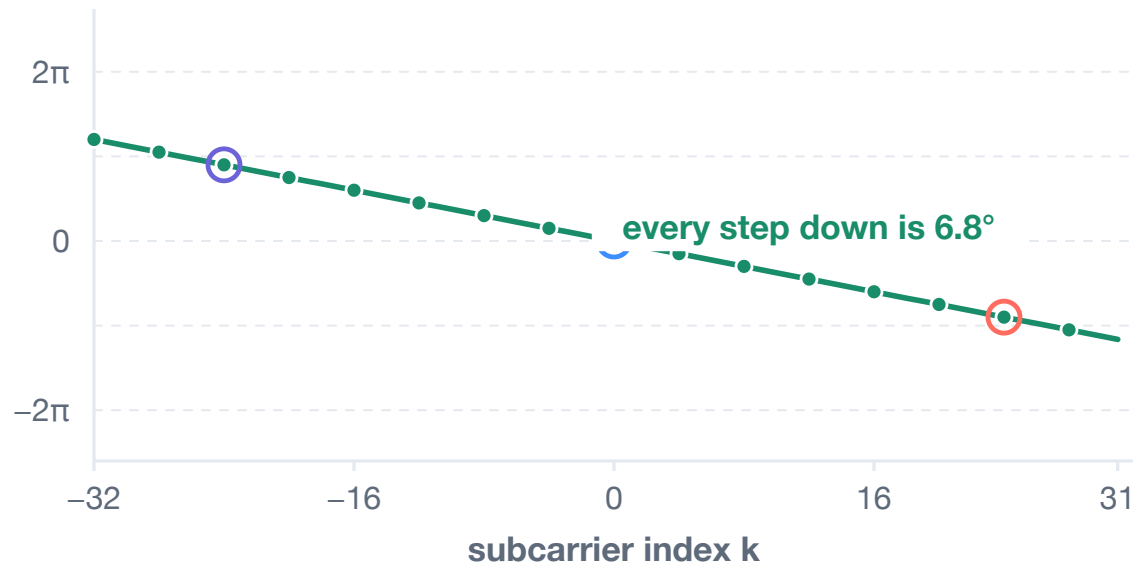
3 Direction

4 Joint

5 Systems

Plot those 64 phases and they fall on a straight line

THE SAME 64 PHASES, PLOTTED AGAINST THE SUBCARRIER INDEX



$$\text{slope} = -2\pi \cdot \Delta f \cdot \tau$$

the same step, taken 64 times

measured step 6.8° per subcarrier

divide by $2\pi \Delta f$ $\tau = 60 \text{ ns}$

times c $d = 18 \text{ m}$

A late FFT window tilts the line too,
so this gives delays apart, not distance.

Drawn unwrapped. The absolute distance already came from two-way ranging; what these 64 numbers add is how many paths arrived and which of them was first.

The slope of the line is the delay.

1 Strength

2 Time

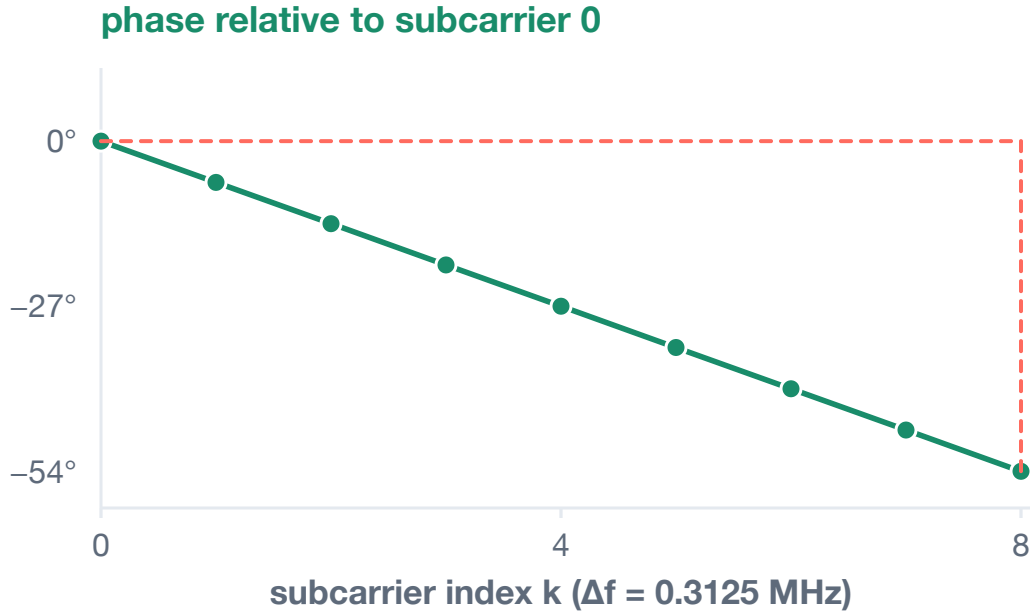
3 Direction

4 Joint

5 Systems

How do we read a delay from the phase slope?

ONE PATH · KNOWN TIMING REFERENCE · UNWRAPPED PHASE



1 Read two phases

$$\phi_0 = 0^\circ \text{ and } \phi_8 = -54^\circ$$

2 Read the slope

$$\Delta\phi / \Delta F = -54^\circ / 2.5 \text{ MHz}$$

3 Solve for the delay

$$\tau = -\Delta\phi / (360^\circ \times \Delta F)$$

$$= 54 / (360 \times 2.5 \times 10^6) \text{ s}$$

$$= 60 \text{ ns (one-way distance } c\tau = 18 \text{ m)}$$

$\Delta F = 8 \times \Delta f = 2.5$ MHz is the gap between the two selected tones. A constant phase offset cancels. Assume the receiver timing offset has been removed.

A 54° phase drop across 2.5 MHz means a 60 ns delay.

1 Strength

2 Time

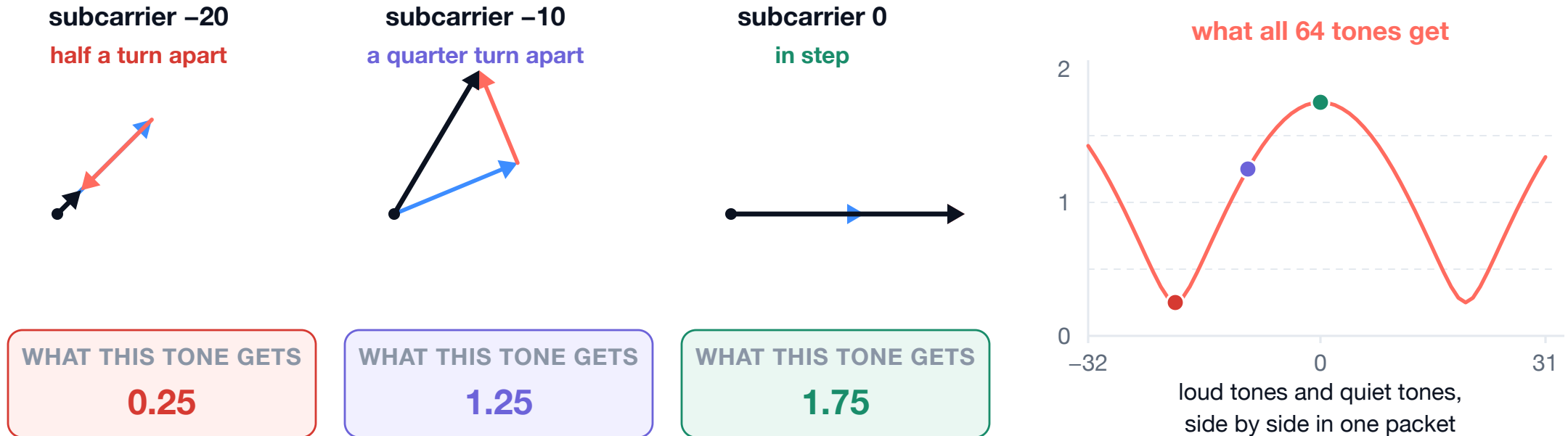
3 Direction

4 Joint

5 Systems

What do two paths do to those 64 numbers?

TWO PATHS: ONE AT 20 ns, ONE AT 100 ns AND THREE QUARTERS AS STRONG



This is slide 18 again, once per subcarrier: blue is the first path, coral the second, dark their sum. The two are further apart on every step up the band, so the sum rises and falls across it.

Every subcarrier adds its own two copies: some come out loud, some nearly vanish.

1 Strength

2 Time

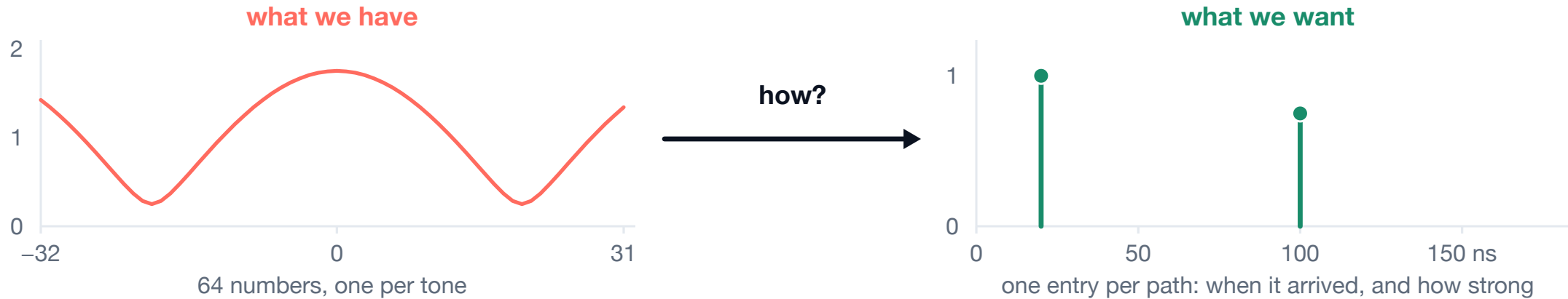
3 Direction

4 Joint

5 Systems

Two paths. So what are we estimating now?

THE MEASUREMENT IS STILL 64 NUMBERS. THE UNKNOWN IS NOT ONE NUMBER ANY MORE.



HOW MANY PATHS

K — unknown

WHEN EACH ARRIVED

$\tau_1 \dots \tau_K$

HOW STRONG EACH IS

$a_1 \dots a_K$

And of that list the one we actually want is the earliest: the direct path is the one whose delay is the distance, and everything after it is a longer way round the room.

K delays, K strengths, and K itself — too many to estimate together. So what now?

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

A path at 20 ns and a path at 100 ns: two waves, two pitches

ONE PATH EACH. THE SAME 64 TONES, THE SAME RULE — ONLY THE DELAY DIFFERS

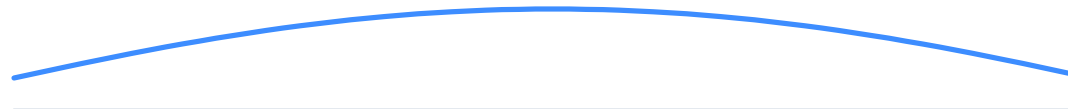
a path at 20 ns (6 m)

the phase steps 2.25° from one tone to the next

the 64 phases as clock hands



the same phases as a wave along k



-32 0 31
0.4 turns of phase across the band = 0.4 cycles of wave

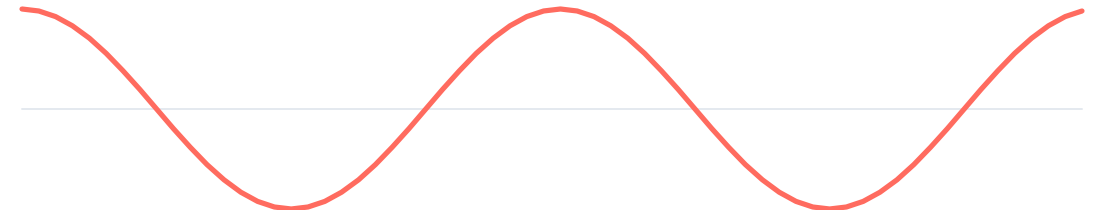
a path at 100 ns (30 m)

the phase steps 11.25° from one tone to the next

the 64 phases as clock hands



the same phases as a wave along k



-32 0 31
2.0 turns of phase across the band = 2.0 cycles of wave

Both phases are a pure sine wave along the subcarrier index, because both are straight lines of phase. The 100 ns path steps five times further per tone, so its wave has five times the frequency. The delay is the pitch.

Both are sine waves along k. The 100 ns path has five times the frequency.

1 Strength

2 Time

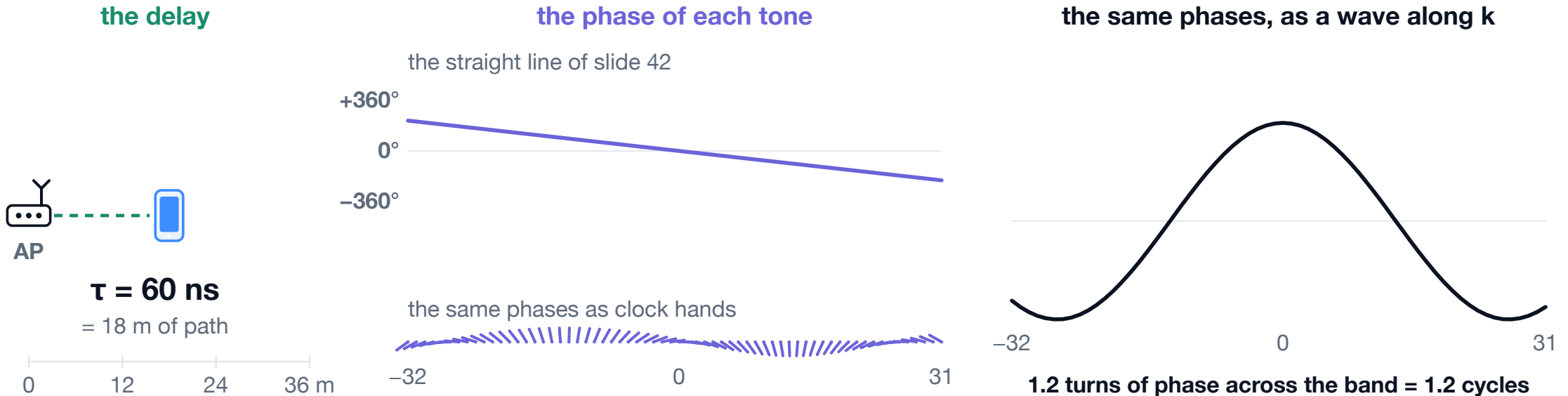
3 Direction

4 Joint

5 Systems

Change the delay, and watch the phase and the wave follow

THE DELAY CHANGES ON THE LEFT, THE PHASES FOLLOW IN THE MIDDLE, THE WAVE FOLLOWS ON THE RIGHT



Animated: the phone walks out to 36 m and back, so the delay sweeps 0 to 120 ns. The line tilts, the hands twist, and the wave speeds up — one full turn of phase across the band is one cycle of wave. Frozen at 60 ns for print.

A longer path is a steeper line of phase, is a faster wave along k. Delay is frequency.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

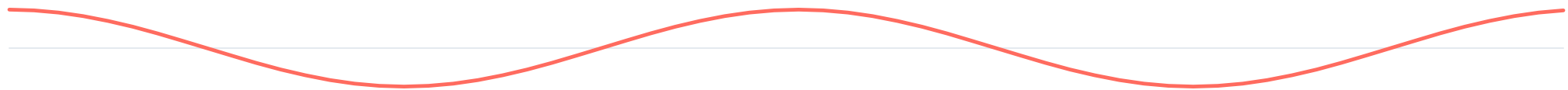
Not a line. Then what is it?

H_k DRAWN AS A WAVE ALONG THE SUBCARRIER INDEX: ONE PATH ALONE, THE OTHER ALONE, AND THE TWO ADDED

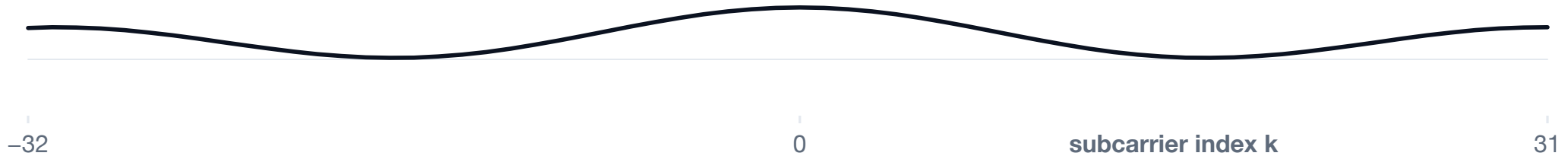
the 20 ns path alone (6 m): it turns 0.4 of a circle across the band — a slow wave



the 100 ns path alone (30 m), three quarters as strong: 2 full circles — five times faster



the two added — this is what we measure. not a line, but not a mess: slow + fast



One path's phase being a straight line is the same as saying H_k turns at a steady rate along k — a pure oscillation whose rate is the delay. Two paths give two rates, added. Delay is a pitch.

Two paths do not make a mess. They make one slow oscillation plus one fast one.

1 Strength

2 Time

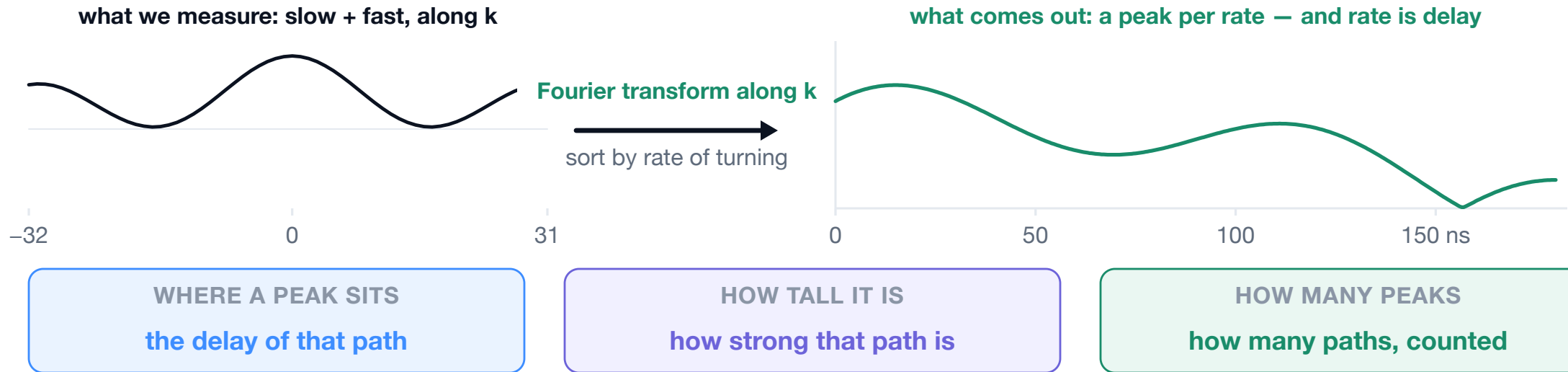
3 Direction

4 Joint

5 Systems

A slow wave plus a fast wave: how do we pull them apart?

A MIXTURE OF PURE OSCILLATIONS HAS ONE CLASSIC UNMIXER, AND THE STUDENTS OF LECTURE 3 ALREADY OWN IT



it works for two reasons: the transform is linear, so the mixture's spectrum is the two spectra added; and a pure oscillation has a spectrum that is a single peak
over 64 equally spaced tones it is one inverse FFT — the receiver already runs one per symbol, so it is free

Nothing is estimated jointly. The three kinds of unknown from slide 45 land in three different places on one picture, and the number of paths is the number of peaks — no one had to guess it first.

The Fourier transform sorts a mixture by rate. Rate is delay, so it sorts the paths by delay.

1 Strength

2 Time

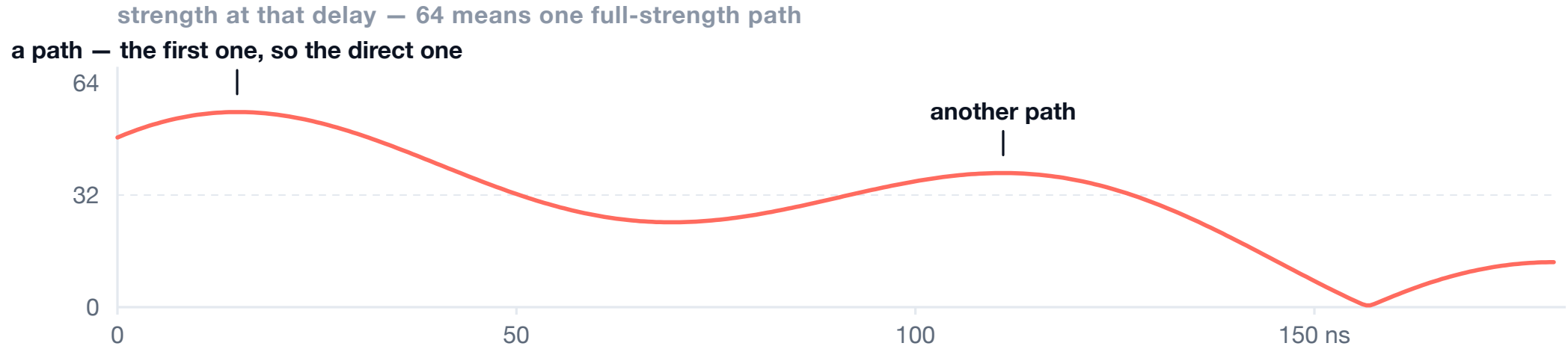
3 Direction

4 Joint

5 Systems

Run it: two peaks, and nobody said how many

THE 64 MEASURED H_k OF THE TWO-PATH CHANNEL, FOURIER-TRANSFORMED ALONG k , DRAWN OVER 0 TO 180 ns



two peaks — so two paths, and nobody said how many. the first peak's delay is the distance.

we read where the peaks are, not where the curve is zero: with two paths present the skirts never reach zero

physically, this curve is the pattern of echoes one short click would have come back with

The peaks sit near 20 and 100 ns, leaning towards each other a little because they are only 80 ns apart on a transform whose peaks are 50 ns wide — the next slide asks how close two can get. This is lecture 4's angle scan, run over delay.

How many, when, how strong: all read off one picture, none estimated jointly.

1 Strength

2 Time

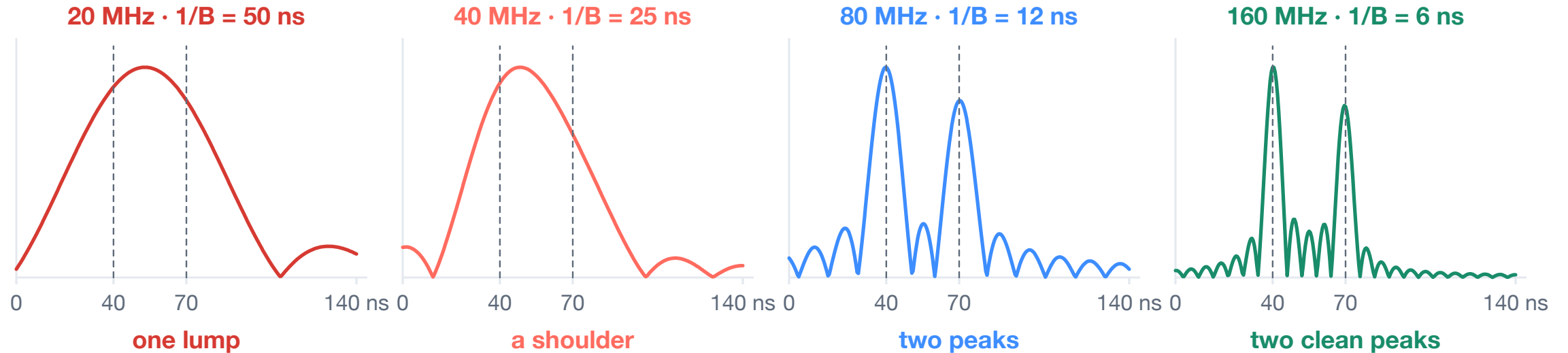
3 Direction

4 Joint

5 Systems

How close can two paths be?

TWO PATHS 30 ns APART, AT FOUR BANDWIDTHS. TWO PATHS SEPARATE ONLY WHEN THEY DIFFER BY A FULL TURN ACROSS THE BAND



Wi-Fi at 20 MHz cannot tell 9 m of extra path from the direct one. At 160 MHz it can.

One full turn across the band is $1 / B$: 50 ns at 20 MHz, 6 ns at 160. Closer, one peak.

1 Strength

2 Time

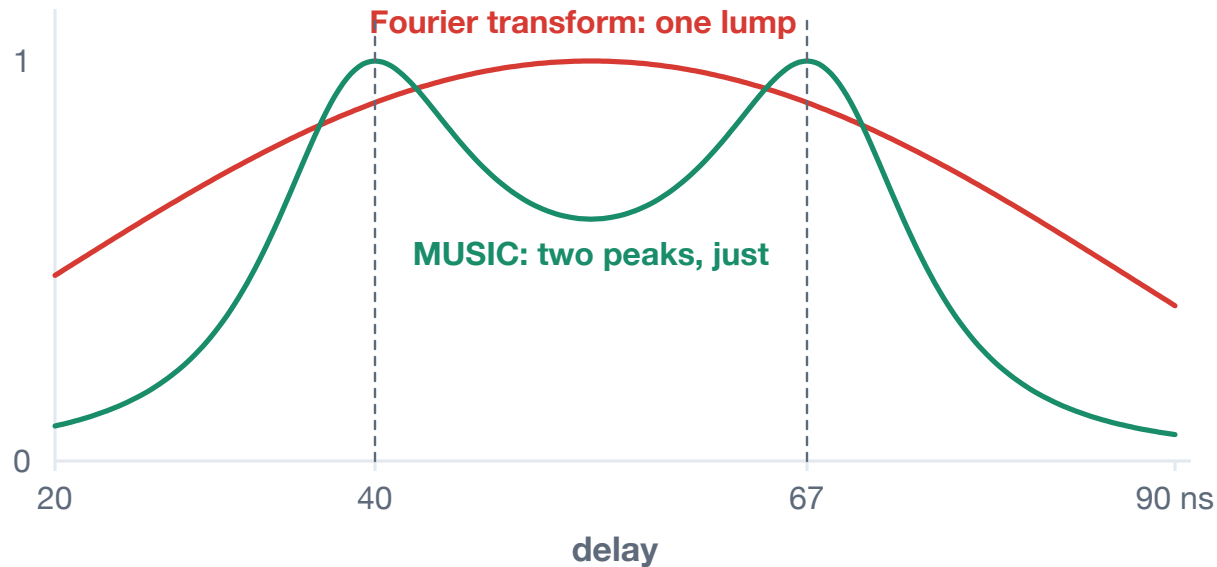
3 Direction

4 Joint

5 Systems

Can we do better than $1/B$?

TWO PATHS 27 ns (8 m) APART AT 20 MHz — THE FOURIER TRANSFORM AGAINST MUSIC, ON THE SAME 64 NUMBERS



what it is

not a transform but a fit: assume the 64 numbers are exactly K pure oscillations plus noise, then find the K rates that leave nothing but noise behind

what you get

peaks about half as wide as the transform's, so two paths it merges just come apart. The price: K has to be given, and it needs several measurements

what it is for

picking the direct path out of the reflections on a narrow Wi-Fi channel. SpotFi runs it over delay and angle at once; Part 4 combines both

MUSIC is MULTiple Signal Classification (Schmidt, 1986), drawn here with a modest noise level. A real gain, not a miracle: about half the width, and the measured curves in the 2023 slides merge below about 5 m at 20 MHz.

MUSIC trades an assumption — how many paths — for peaks about half as wide.

1 Strength

2 Time

3 Direction

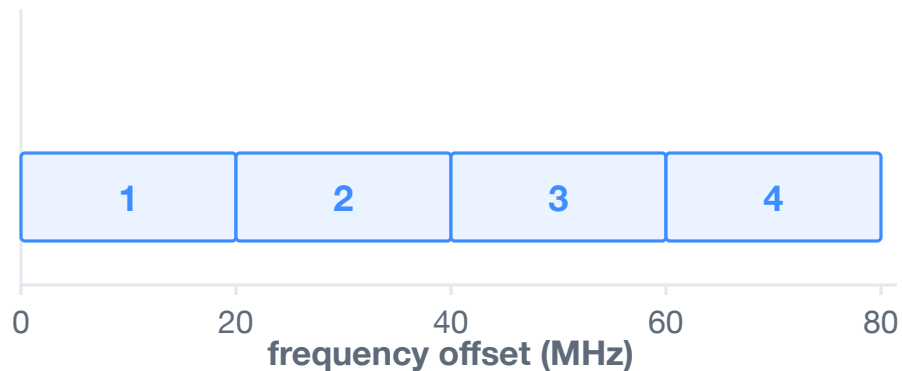
4 Joint

5 Systems

How can we increase the bandwidth?

TWO WAYS TO COVER A WIDER FREQUENCY RANGE OVER TIME

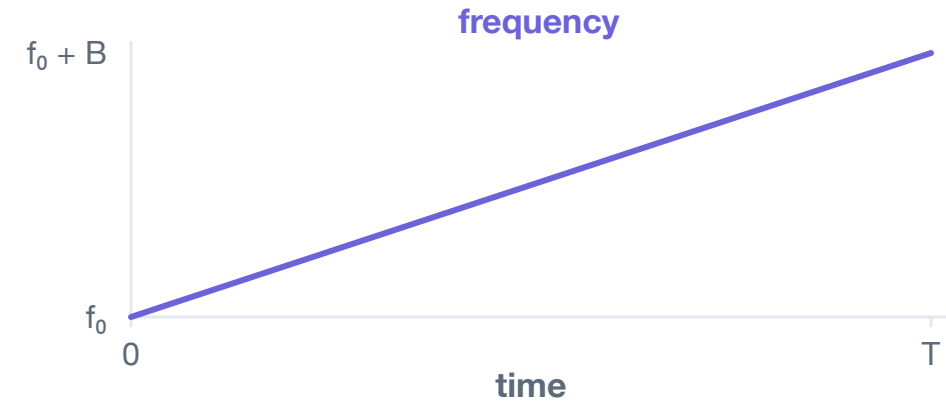
Channel stitching



Measure neighbouring channels in turn.

Align their phases to form one wider band.

FMCW: a continuous frequency sweep



Transmit one tone whose frequency rises.

Tune an oscillator smoothly across B.

Stitching combines separate channel measurements. FMCW (frequency-modulated continuous wave) sweeps through the band in one chirp, which is convenient to generate with a tunable oscillator.

Hop across channels, or sweep continuously: both cover a wider band.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

Which way did the signal come from?

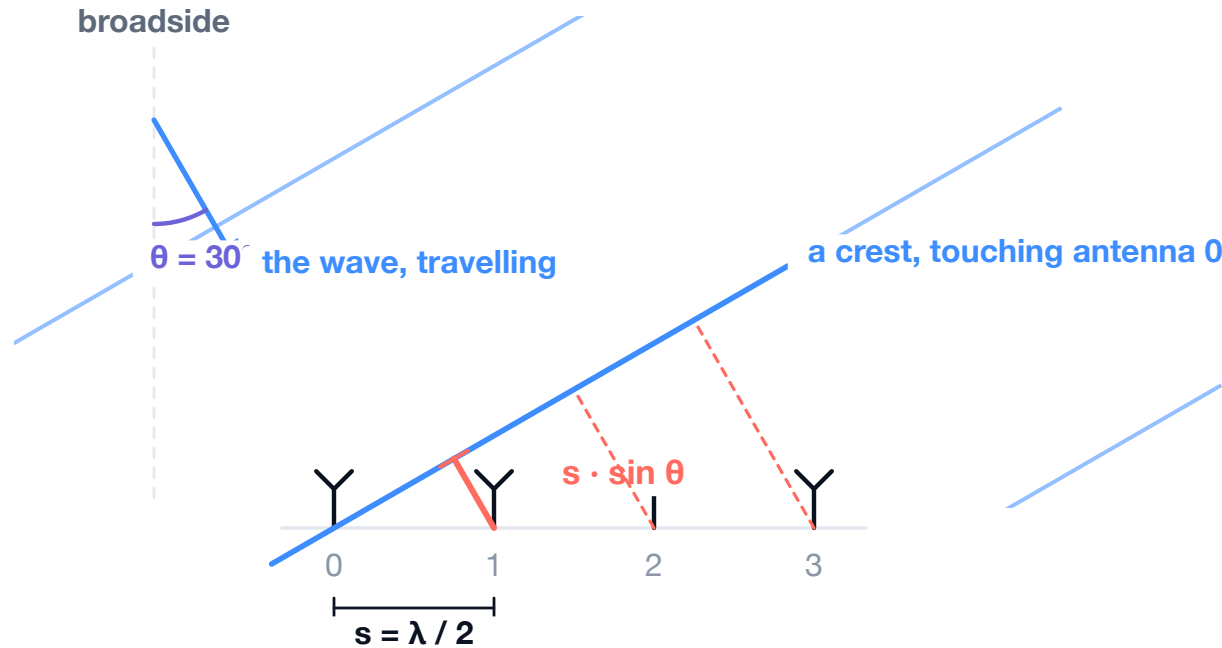
03

Direction from phase

Lecture 4 showed that a wave arriving at an angle reaches each antenna a little later, so every antenna measures a slightly different phase. This part reads those phases backwards: the step between neighbours gives the angle, the angle is a line on the floor plan, and the number of antennas sets how sharp that line is.

A wave from the side reaches each antenna a little later

ONE PATH, FOUR ANTENNAS HALF A WAVELENGTH APART, THE WAVE ARRIVING 30° OFF BROADSIDE



flat crests

parallel crests one wavelength apart, coming in 30° off broadside

antenna 0 hears it first

a crest touches antenna 0 before it reaches any of the others

each neighbour is $s \cdot \sin \theta$ further on

antenna 1 waits for $s \cdot \sin \theta$ more travel; antenna 2 for twice that, antenna 3 for three times

the same wave, a little later each time

later by $s \cdot \sin \theta / c$ per step; with $s = \lambda / 2$ and 30° , a quarter of a period

Animated: the crests sweep across the array and the dots light up in order, 0 then 1 then 2 then 3, a quarter of a period apart. One path only; multipath comes later.

Off broadside, each antenna is $s \cdot \sin \theta$ further along the wave than the one before it.

1 Strength

2 Time

3 Direction

4 Joint

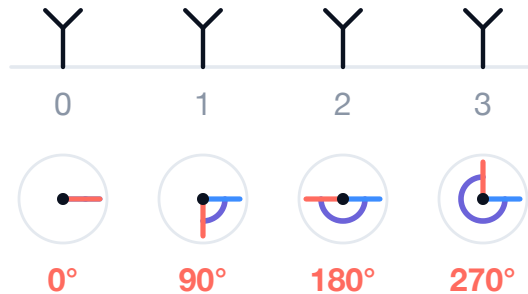
5 Systems

What does the extra distance do to the phase?

THE SAME FOUR ANTENNAS: $s = \lambda / 2$, $\theta = 30^\circ$, SO EACH NEIGHBOUR IS $\lambda / 4$ FURTHER ALONG — A QUARTER TURN OF PHASE

$$\Delta\phi = 2\pi \cdot s \sin \theta / \lambda$$

extra distance, in wavelengths, times 2π
with $s = \lambda / 2$ and $\theta = 30^\circ$: $\Delta\phi = 90^\circ$



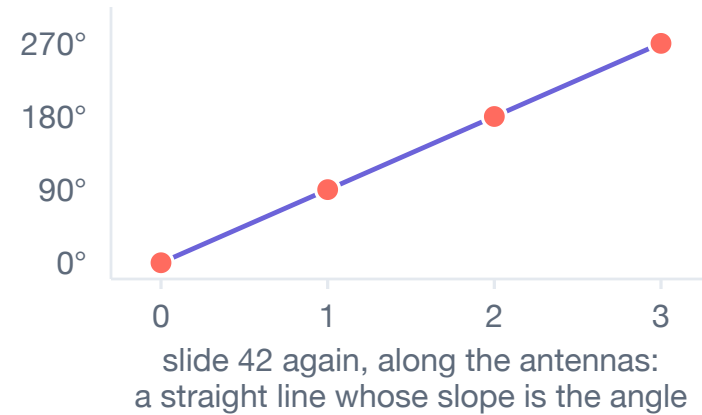
the phase each antenna measures

everything Part 2 said about the tones holds along the antennas, with the angle in place of the delay

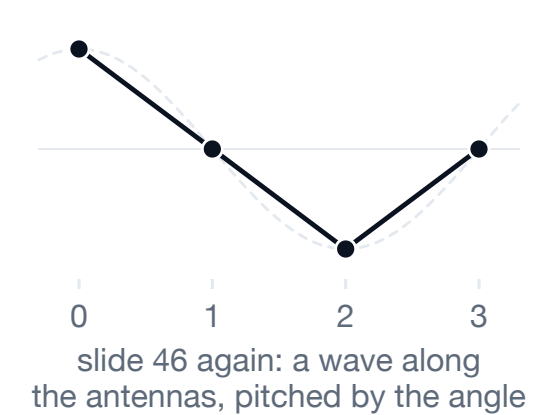
antenna index $\leftrightarrow$ subcarrier index $\cdot$ angle $\leftrightarrow$ delay $\cdot$ more antennas $\leftrightarrow$ more bandwidth

The extra distance of the last slide, divided by the wavelength and times 2π , is a phase. It is the same extra distance between every pair of neighbours, so the phase climbs in equal steps — a straight line against the antenna index.

the phase against the antenna index



the same four, as a wave



Along the antennas the phase is a straight line again, and its slope is the angle.

1 Strength

2 Time

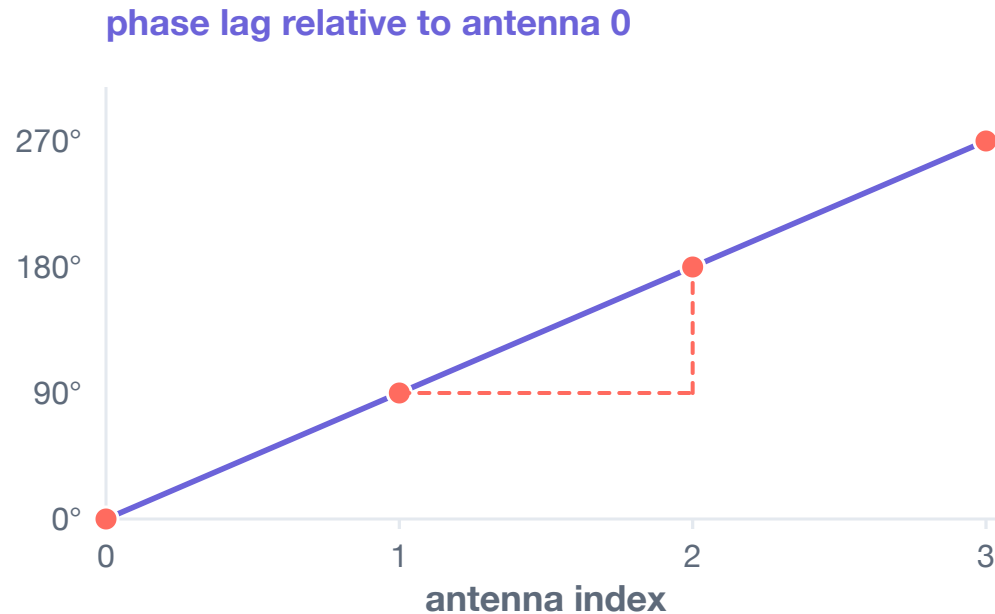
3 Direction

4 Joint

5 Systems

How do we read an angle from the phase slope?

ONE PATH · ANTENNA SPACING $s = \lambda / 2$ · ANGLE θ MEASURED FROM BROADSIDE



1 Read the antenna phases

0°, -90°, -180°, -270° (unwrapped)

2 Find the lag per antenna step

$$\Delta\phi = (270^\circ - 0^\circ) / 3 = 90^\circ = \pi/2$$

3 Convert the lag to an angle

$$\sin \theta = \lambda \Delta\phi / (2\pi s) = (\pi/2) / \pi = 1/2$$

$$\theta = \arcsin(1/2) = 30^\circ$$

The graph uses positive phase lag: how far each antenna falls behind antenna 0. Assume calibrated antennas. With $s = \lambda/2$, $\Delta\phi = \pi \sin \theta$, so the slope gives $\sin \theta$.

A 90° lag per antenna step means an arrival angle of 30° from broadside.

1 Strength

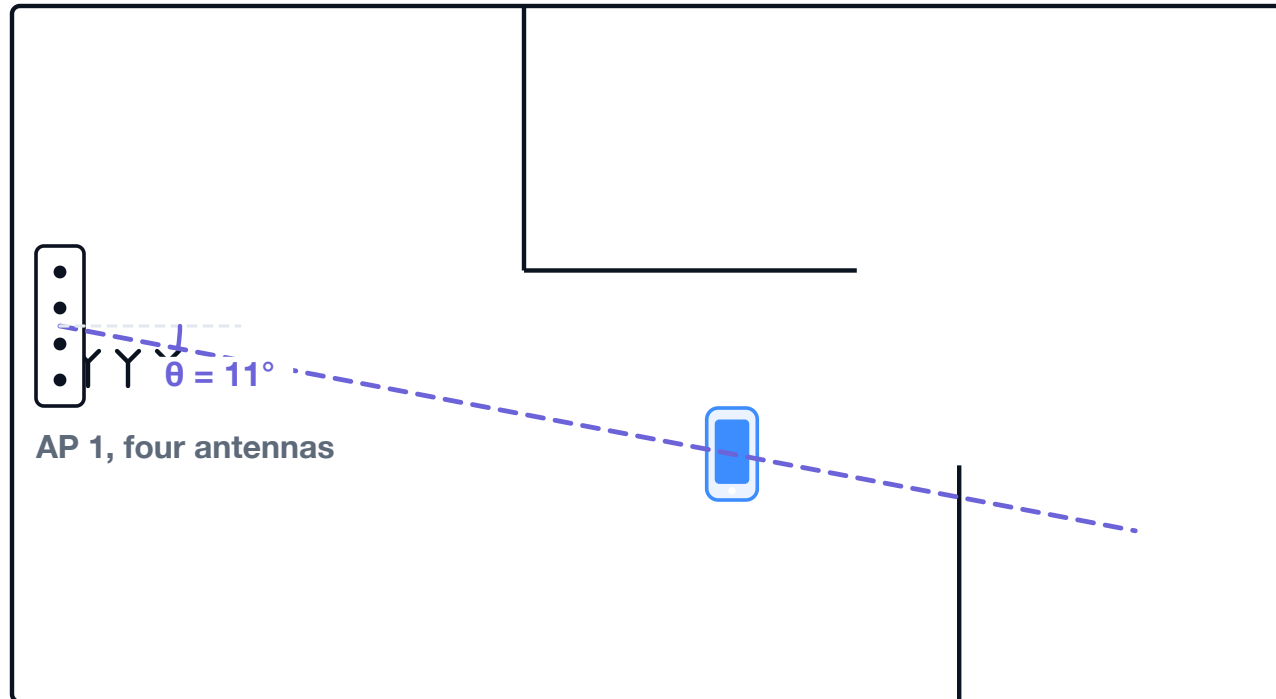
2 Time

3 Direction

4 Joint

5 Systems

One angle gives one line



AP 1 measures one angle

the phone is somewhere on the line
leaving AP 1 at that angle

One angle, and the phone is anywhere on the line.

The AP does not know the distance.

Compare Part 1: one RSSI gave a circle.
A line and a circle are the two shapes on the plan.

One anchor with an array: the phone is on a line.

1 Strength

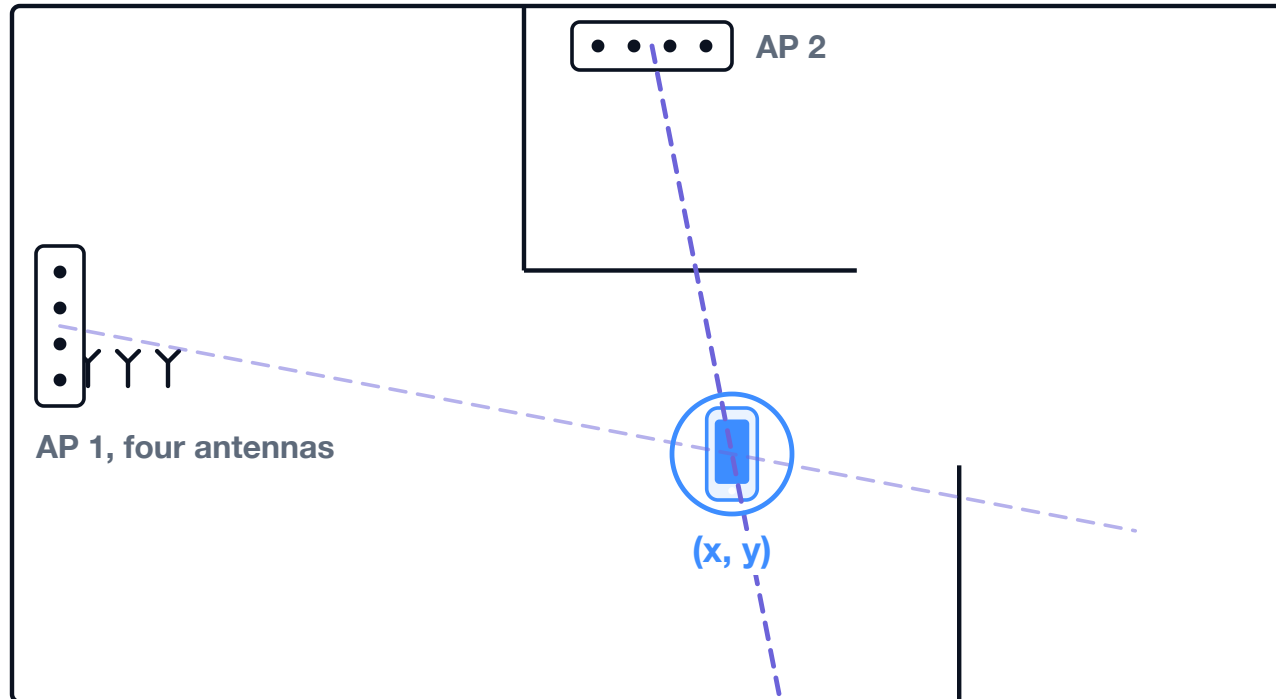
2 Time

3 Direction

4 Joint

5 Systems

Two lines cross at one point



two APs, two angles

two lines cross at exactly one point

triangulation

two directions from two known points fix one position

No distance was measured. Two anchors are enough, where trilateration needed three.

Two arrays fix the position without any distance. This is triangulation.

1 Strength

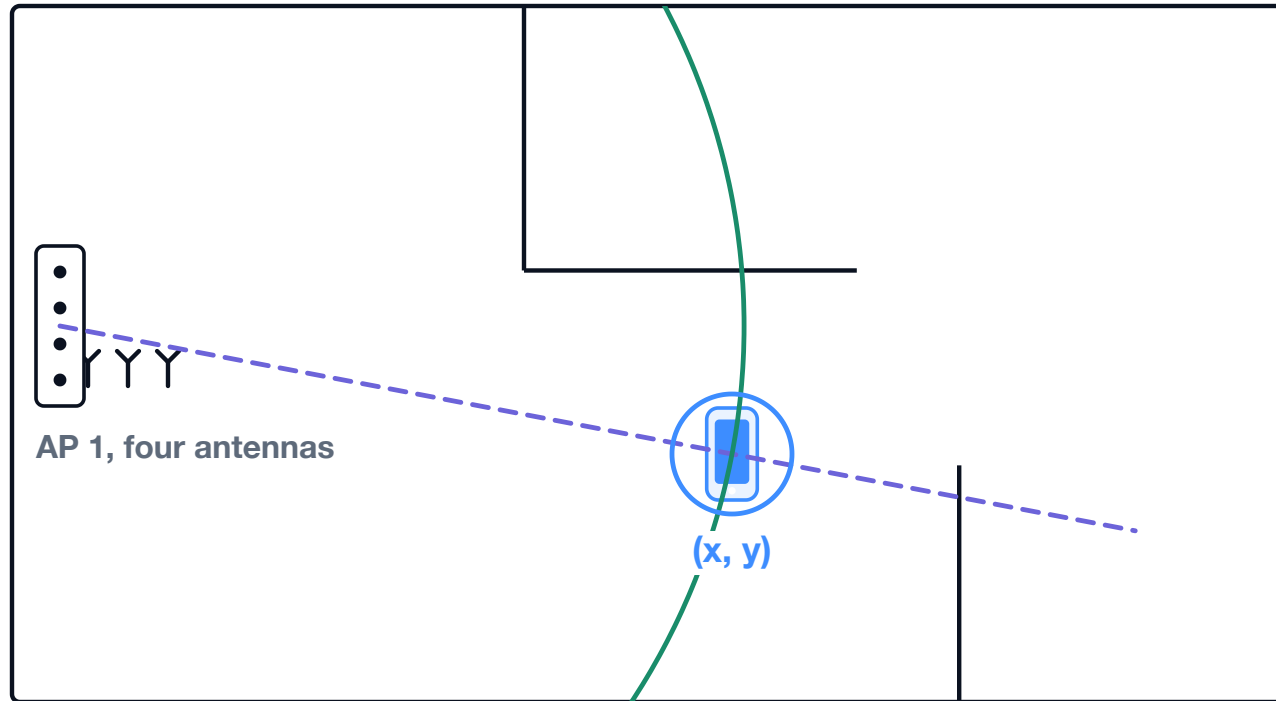
2 Time

3 Direction

4 Joint

5 Systems

One anchor, angle and distance together



the array gives

$$\theta = 11^\circ$$

two-way ranging gives

$$d = 8.6 \text{ m}$$

one anchor is enough

a line and a circle from the same AP cross once

This is how a phone finds a UWB tag: the tag's distance and direction, from the phone alone.

Angle plus distance from one anchor: polar coordinates of the phone.

1 Strength

2 Time

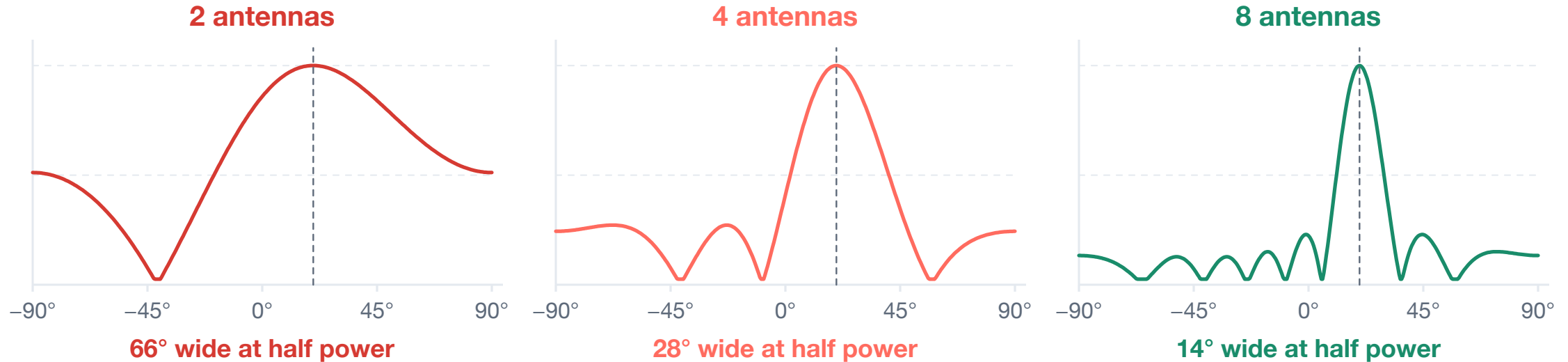
3 Direction

4 Joint

5 Systems

How sharply does an array see an angle?

DELAY-AND-SUM SCAN, ONE SOURCE AT 20°, HALF-WAVELENGTH SPACING



Twice the antennas, half the width: beam width $\approx \lambda / (N s)$ radians once N is more than a few.

Beam width falls as $1 / N$: 66° for two antennas, 14° for eight.

1 Strength

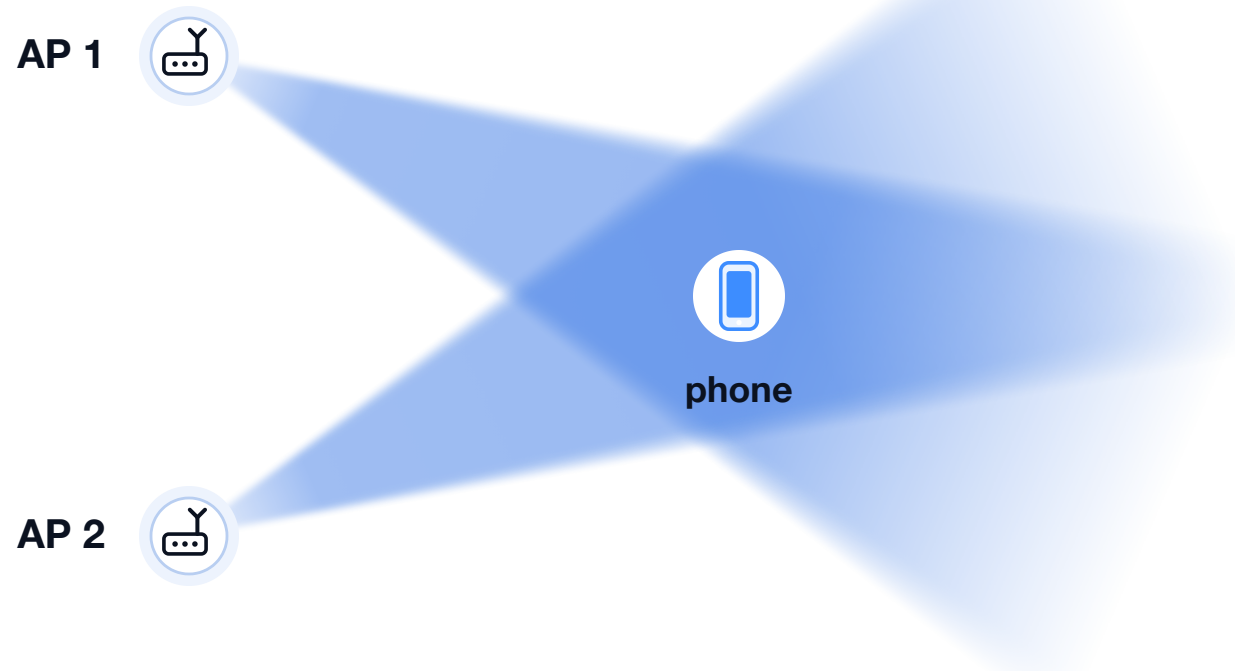
2 Time

3 Direction

4 Joint

5 Systems

Where do two beams put the phone?



BOTH APs

28°

beamwidth

Two translucent beams.

A darker overlap.

The phone is roughly here.

The APs and phone stay in the same positions. The beams illustrate angular uncertainty, with darker color where they overlap.

The phone is approximately where the two beams overlap.

1 Strength

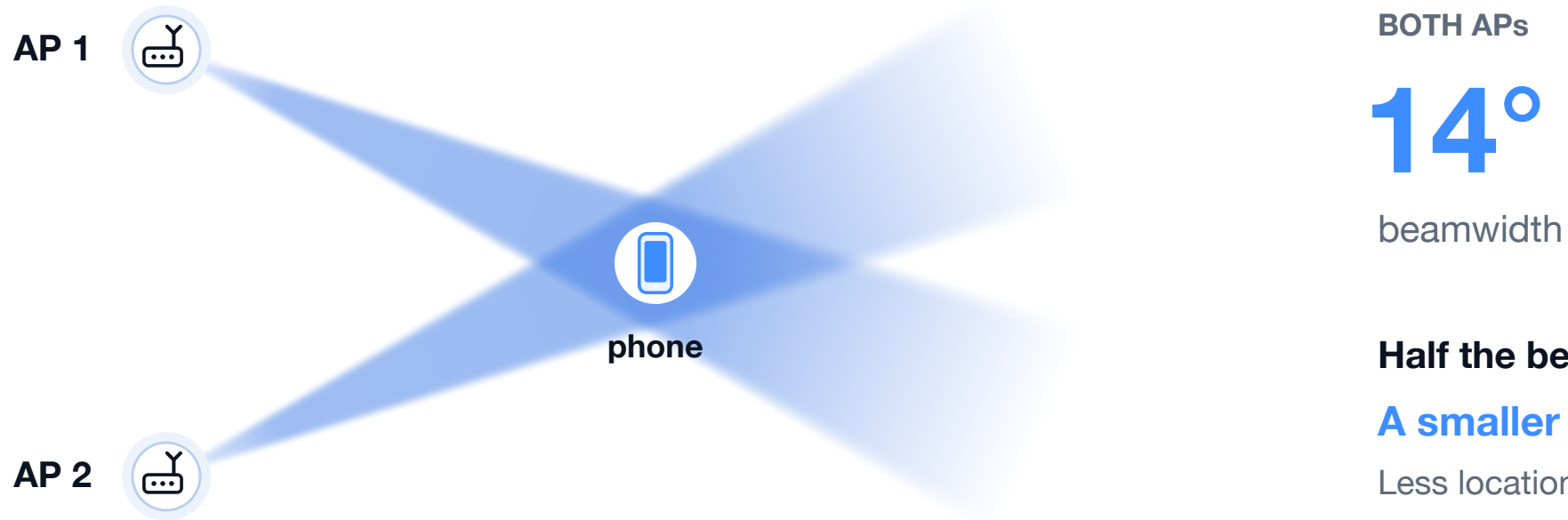
2 Time

3 Direction

4 Joint

5 Systems

Narrower beams leave a smaller region



Half the beamwidth.

A smaller overlap.

Less location uncertainty.

The APs and phone stay in the same positions. The beams illustrate angular uncertainty, with darker color where they overlap.

More antennas narrow the beams and shrink the location uncertainty.

1 Strength

2 Time

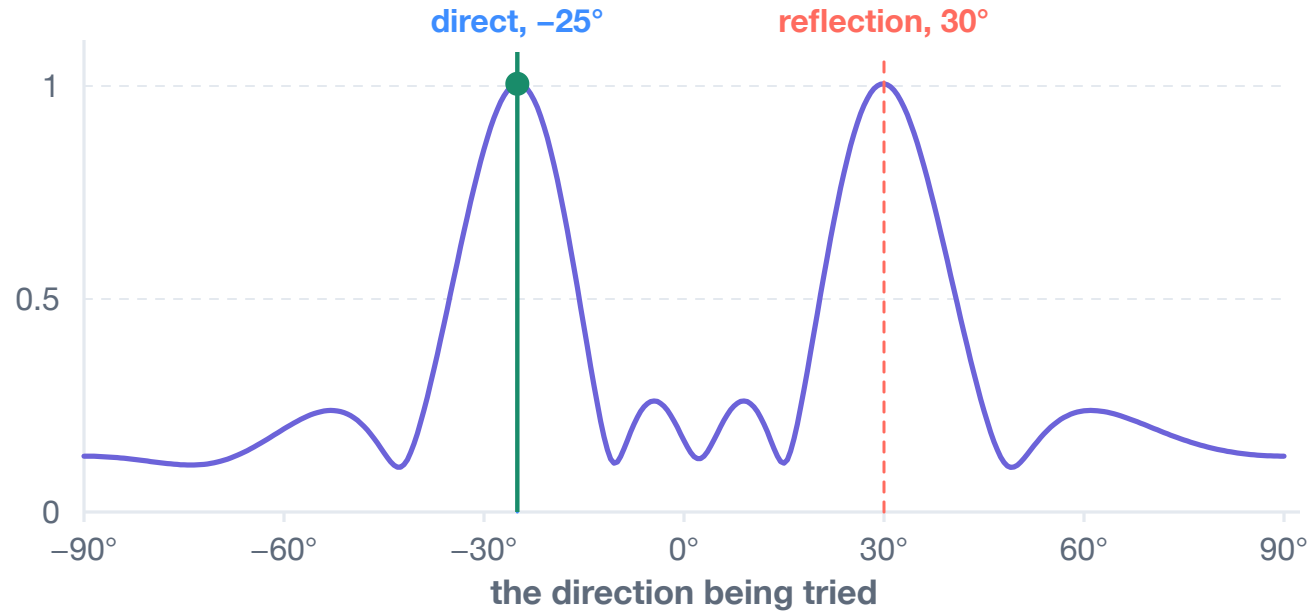
3 Direction

4 Joint

5 Systems

Two paths arriving at one array

EIGHT ANTENNAS, TWO ARRIVALS: THE DIRECT PATH AT -25° AND A REFLECTION AT 30°



two arrivals

two phase lines added: two waves, two pitches

the scan finds both

one peak per path, each about 13° wide

the same trick as Part 2

a Fourier transform along the antennas

The spectrum separates paths by angle, as the delay profile did by delay.

1 Strength

2 Time

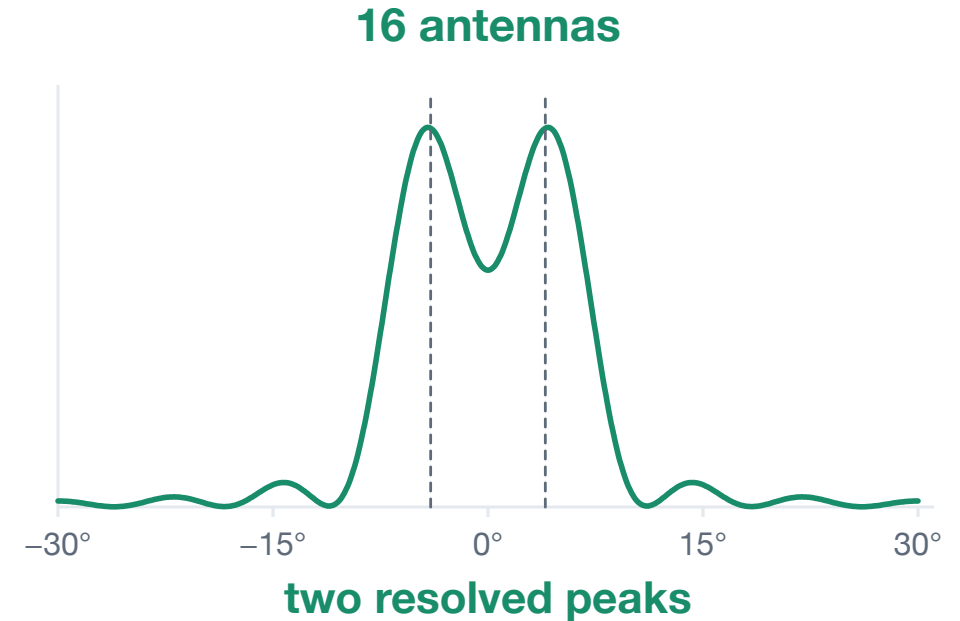
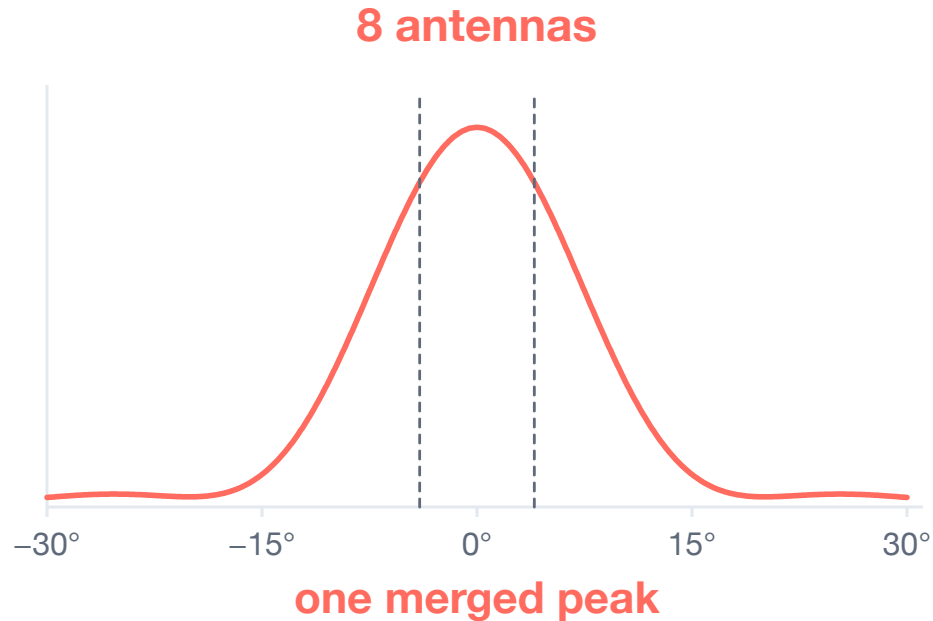
3 Direction

4 Joint

5 Systems

What if two arrival angles are too close?

THE SAME TWO ARRIVALS: -4° AND $+4^\circ$, ONLY 8° APART · ANTENNA SPACING $\lambda / 2$



Computed conventional beam scans for two equal-power, decorrelated arrivals. Dashed lines mark the true angles. Each plot is normalized to its maximum.

As bandwidth sharpens delay, a larger array sharpens angle. Close paths can merge.

1 Strength

2 Time

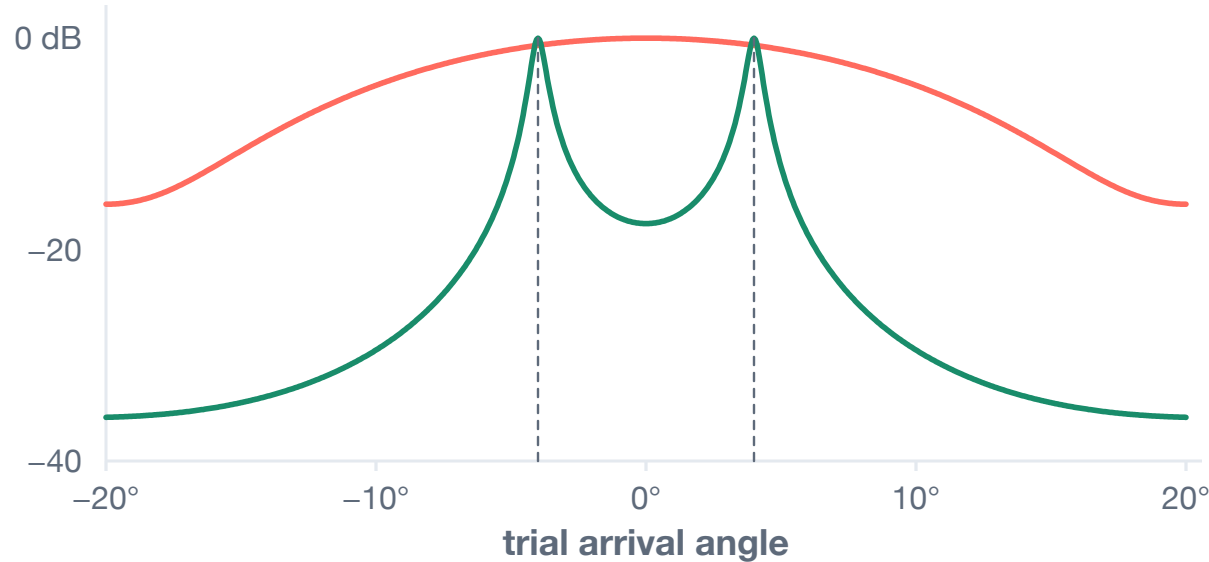
3 Direction

4 Joint

5 Systems

Can MUSIC separate the merged angles?

SAME EIGHT ANTENNAS, SAME TWO ANGLES · IDEAL SUBSPACE EXAMPLE



Conventional scan

The two broad responses merge.

MUSIC

Test angles against the noise subspace.

Two peaks: -4° and $+4^\circ$

The same idea as MUSIC for delay.

Ideal calibrated array with a known two-path signal subspace. In practice, MUSIC needs sufficient SNR and samples; coherent multipath requires smoothing or other decorrelation.

MUSIC can separate two angles within one conventional beam.

1 Strength

2 Time

3 Direction

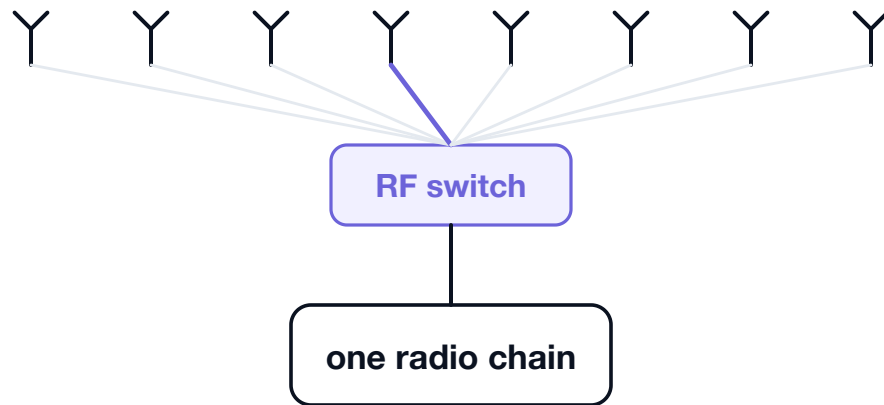
4 Joint

5 Systems

Where do the antennas come from?

A COMMODITY ACCESS POINT HAS THREE OR FOUR RADIO CHAINS. TWO WAYS TO GET MORE ANTENNAS

antenna switching



sample the antennas one after another, within one packet
eight antennas on one chain; the channel must hold still meanwhile

a moving antenna: synthetic aperture



one antenna, measured at eight positions in turn
a robot, a drone, or a hand: the track is the array
again the channel and the source must hold still

More antennas without more radios: switch between them, or move one.

1 Strength

2 Time

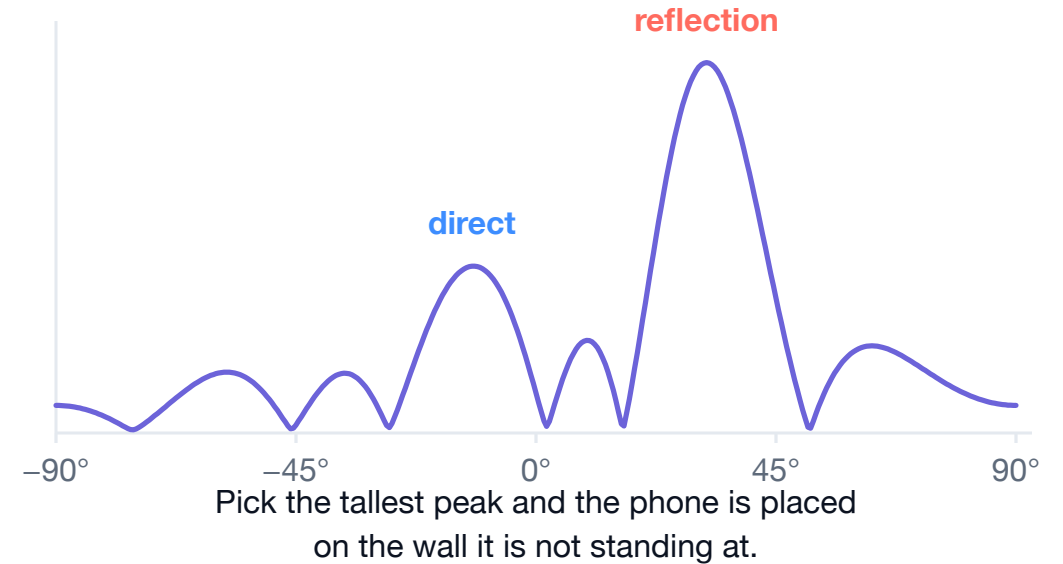
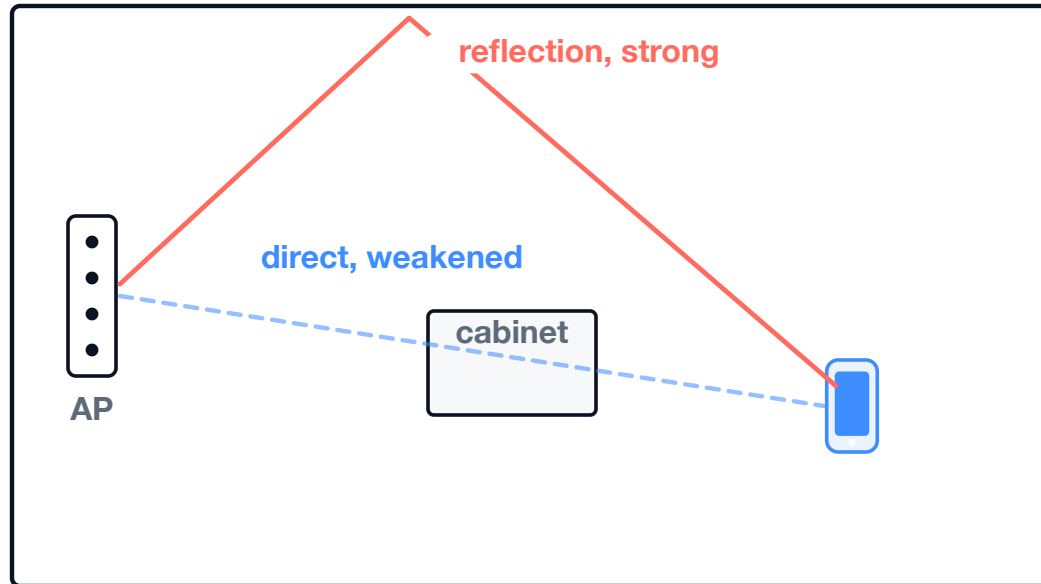
3 Direction

4 Joint

5 Systems

Which peak is the direct path?

THE DIRECT PATH GOES THROUGH A CABINET; THE REFLECTION DOES NOT



The direct path is the one that points at the phone. A wall, a cabinet or a person in the way can make a reflection the stronger arrival.

The strongest peak is not always the direct one.

1 Strength

2 Time

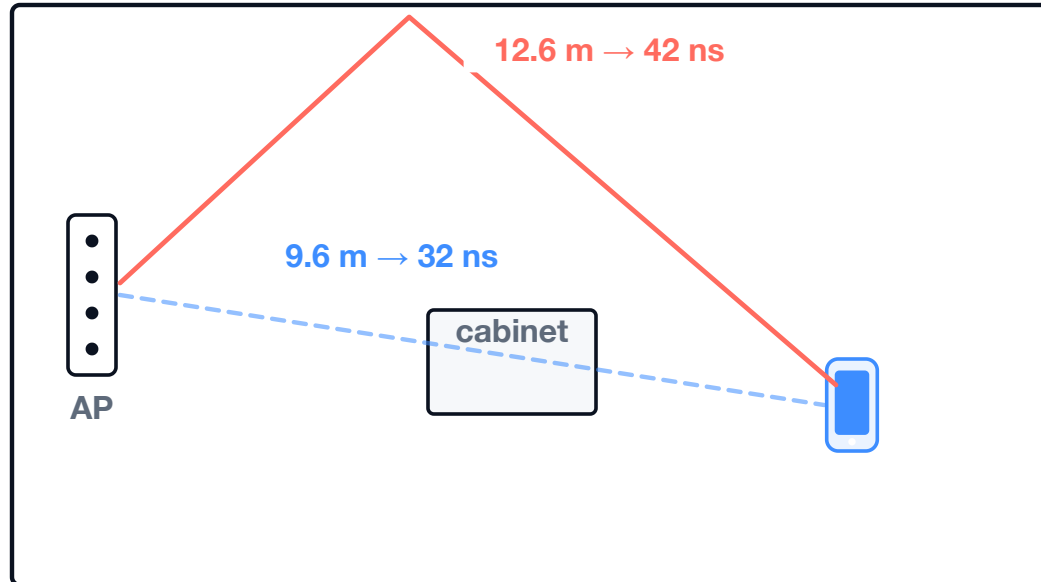
3 Direction

4 Joint

5 Systems

Pick the shortest one

THE SAME ROOM: EVERY PATH HAS AN ANGLE, A DELAY AND A STRENGTH



path	angle	delay	strength
direct	-12°	32 ns	0.45
reflection	32°	42 ns	1.00

the direct path, when visible, arrives first

a reflection travels farther, so it arrives later, every time

So: estimate angle and delay for each path, and take the angle of the earliest one. Strength decides nothing.

When a direct path is detectable, its arrival precedes the reflections.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

Can two dimensions separate more paths?

04

Multidimensional localization

Measure a wideband signal on several antennas. Estimate each path's angle and delay together.

Two kinds of diversity in one measurement

ONE WIDEBAND TRANSMISSION, RECEIVED BY MULTIPLE ANTENNAS

Spatial diversity

Observe the wave at different positions.



Phase across antennas gives angle.

Larger aperture sharpens angle.

Frequency diversity

Observe the channel at different frequencies.



Phase across frequencies gives delay.

Wider bandwidth sharpens delay.

Each antenna measures all the subcarriers in the same packet. Spatial and frequency diversity provide two different ways to distinguish propagation paths.

Keep both dimensions of the measurement, then estimate angle and delay together.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

Each path leaves a pattern across both axes

CSI: A COMPLEX CHANNEL MEASUREMENT FOR EVERY ANTENNA AND EVERY SUBCARRIER

Across frequencies: the delay pattern

	f_0	f_1	f_2	...	f_{K-1}
antenna 0	$h_{0,0}$	$h_{0,1}$	$h_{0,2}$	...	$h_{0,K-1}$
antenna 1	$h_{1,0}$	$h_{1,1}$	$h_{1,2}$	...	$h_{1,K-1}$
antenna 2	$h_{2,0}$	$h_{2,1}$	$h_{2,2}$	...	$h_{2,K-1}$
antenna 3	$h_{3,0}$	$h_{3,1}$	$h_{3,2}$	...	$h_{3,K-1}$

Measure the whole matrix

N antennas $\times$ K subcarriers

One path has two phase slopes

One for delay, one for angle.

Fit both slopes to the same path

Each estimate is a pair (θ, τ) .

Across antennas: the angle pattern

Test candidate (angle, delay) pairs against all entries together. Joint MUSIC can perform this search after smoothing the CSI to separate coherent multipath.

The two coordinates belong to the same path, so there is no separate pairing step.

1 Strength

2 Time

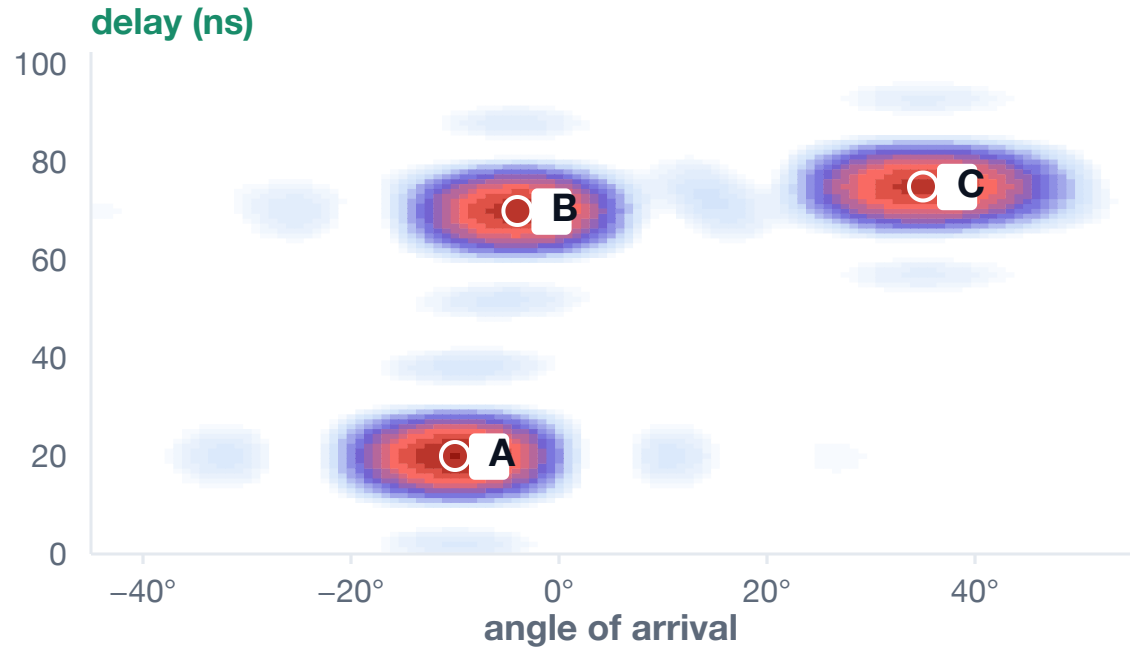
3 Direction

4 Joint

5 Systems

A path hidden on one axis separates on the other

THREE PATHS · EIGHT ANTENNAS · 80 MHz BANDWIDTH · JOINT ANGLE-DELAY SCAN



A: $(-10^\circ, 20 \text{ ns})$

B: $(-4^\circ, 70 \text{ ns})$ C: $(35^\circ, 75 \text{ ns})$

A and B: only 6° apart

Their 50 ns delay gap separates them.

B and C: only 5 ns apart

Their 39° angle gap separates them.

A arrives first: use its angle for localization.

Computed conventional joint scan for ideal decorrelated paths, with a common timing reference. Joint MUSIC uses the same two axes for finer separation. A visible direct path arrives first.

Joint measurements separate more paths and can improve the direct-path angle estimate.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

Adding axes: Doppler and angle of departure

EVERY AXIS THE CHANNEL OFFERS IS ANOTHER WAY TO TELL TWO PATHS APART

angle of arrival

the phase step across the
receive antennas

resolution $\lambda / (N s)$

time of flight

the phase step across the
subcarriers

resolution $1 / B$

Doppler

the phase step from packet to
packet, if anything moves

resolution $1 / T$

angle of departure

the phase step across the
transmit antennas

resolution $\lambda / (M s)$

mD-Track, MobiCom 2019

estimates all four for every path jointly, from commodity Wi-Fi, and tracks a person who carries no device at all
Xie, Xiong, Li and Jamieson, “mD-Track: Leveraging multi-dimensionality for passive indoor Wi-Fi tracking”

More dimensions, better separation of paths. This is also where sensing starts.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

Where does all of this end up?

05

Systems and summary

Strength, time and direction, each with its geometry and its failure. This part puts the real systems on one table, names the two numbers that set every accuracy on it, and points at the lecture that comes next.

What each method reaches

TYPICAL ERRORS, AS REPORTED FOR EACH SYSTEM; INDOORS UNLESS SAID OTHERWISE

system	what it measures	typical error	what it needs
GPS, open sky	one-way time, atomic clocks	≈ 5 m	four satellites in view
RSSI fingerprinting	a vector of strengths	2 – 3 m	a survey of the building
Wi-Fi FTM (RTT)	two-way time, 20–160 MHz	1 – 2 m	802.11mc access points
Wi-Fi CSI, angle + delay	both phase steps	0.3 – 0.5 m	3 to 8 antennas per AP
UWB two-way ranging	two-way time, 500 MHz	≈ 0.1 m	a UWB radio at both ends

Read down the error column against the bandwidth and antenna columns: the accuracy follows the resources, not the method's name.

CSI figures: ArrayTrack (NSDI 2013, eight antennas per AP, median 23 cm) and SpotFi (SIGCOMM 2015, three antennas, median 40 cm). Research systems, not products.

Accuracy tracks bandwidth and antennas, not the cleverness of the method.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

Two resolutions, two resources

THE TWO NUMBERS THAT SET EVERY ROW OF THE LAST TABLE

delay

$$\Delta\tau \approx 1 / B$$

bandwidth sharpens time

20 MHz → 15 m · 500 MHz → 0.6 m

angle

$$\Delta\theta \approx \lambda / (N s)$$

antennas sharpen direction

2 antennas → 66° · 8 antennas → 14°

Strength has no such number.

RSSI is one value, and multipath moves it by more than the distance does. That is why it stops at the room.

Bandwidth sharpens time; antennas sharpen angle. Nothing sharpens strength.

1 Strength

2 Time

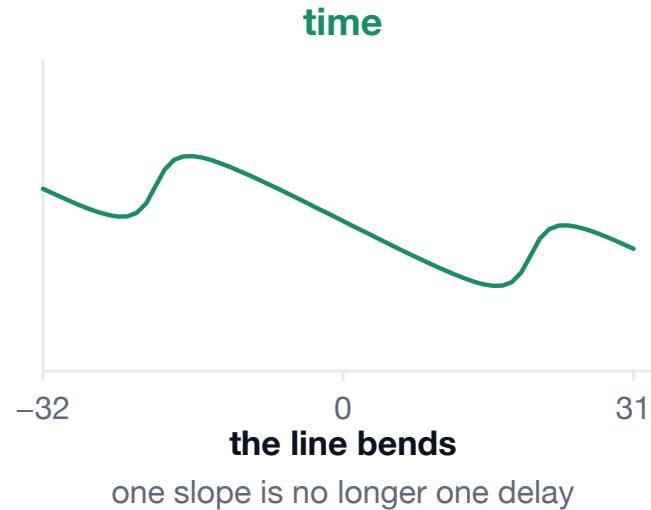
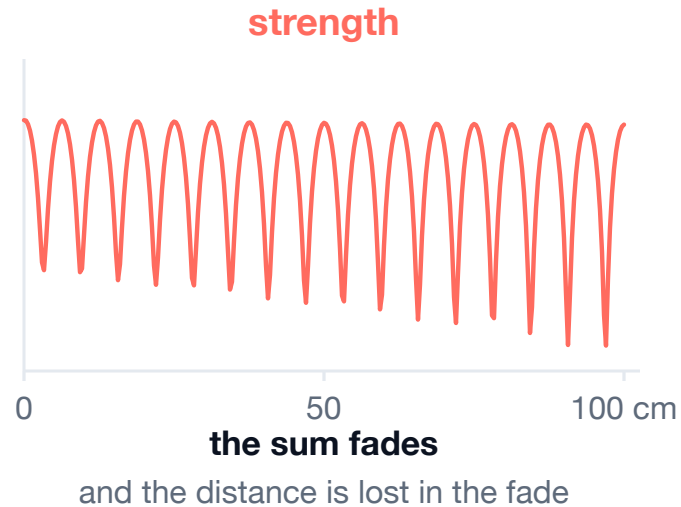
3 Direction

4 Joint

5 Systems

The common enemy

THE SAME TWO PATHS, SEEN THREE WAYS



Every fix is the same fix: measure along enough axes that the paths come apart, then keep the earliest.

Every method fails on multipath, and every fix separates the paths.

1 Strength

2 Time

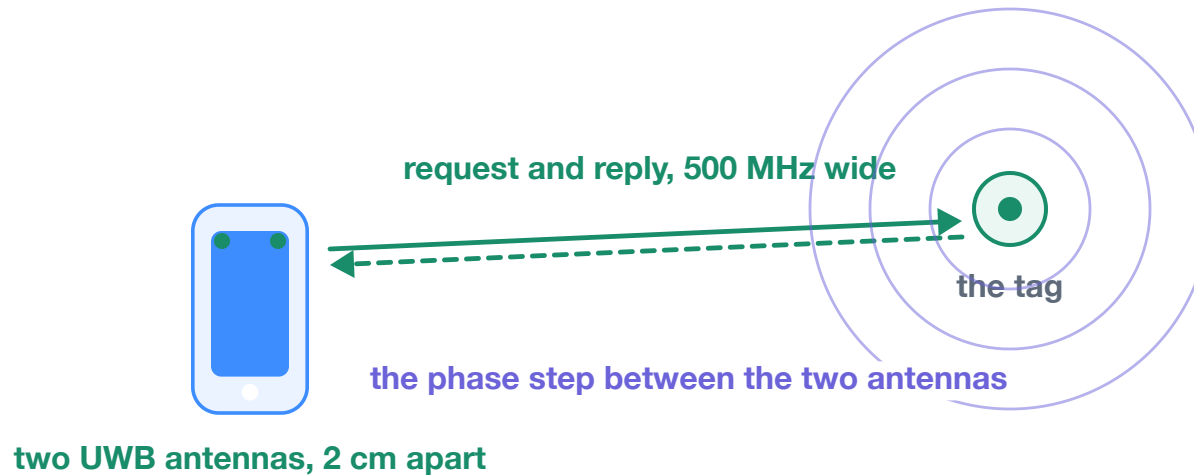
3 Direction

4 Joint

5 Systems

UWB in a phone

A PHONE FINDING A TAG: DISTANCE FROM TWO-WAY RANGING, DIRECTION FROM TWO ANTENNAS



two-way ranging

d to ≈ 10 cm

two antennas

θ to a few degrees

together

an arrow on the screen

Part 3, slide “one anchor, angle and distance together”, built into a phone. Bandwidth bought the distance; the second antenna bought the direction.

At 6.5 GHz the wavelength is 4.6 cm, so two antennas half a wavelength apart fit inside a phone. The same two measurements, at Wi-Fi wavelengths, would not.

Ten centimetres and a direction, from one anchor, because of 500 MHz.

1 Strength

2 Time

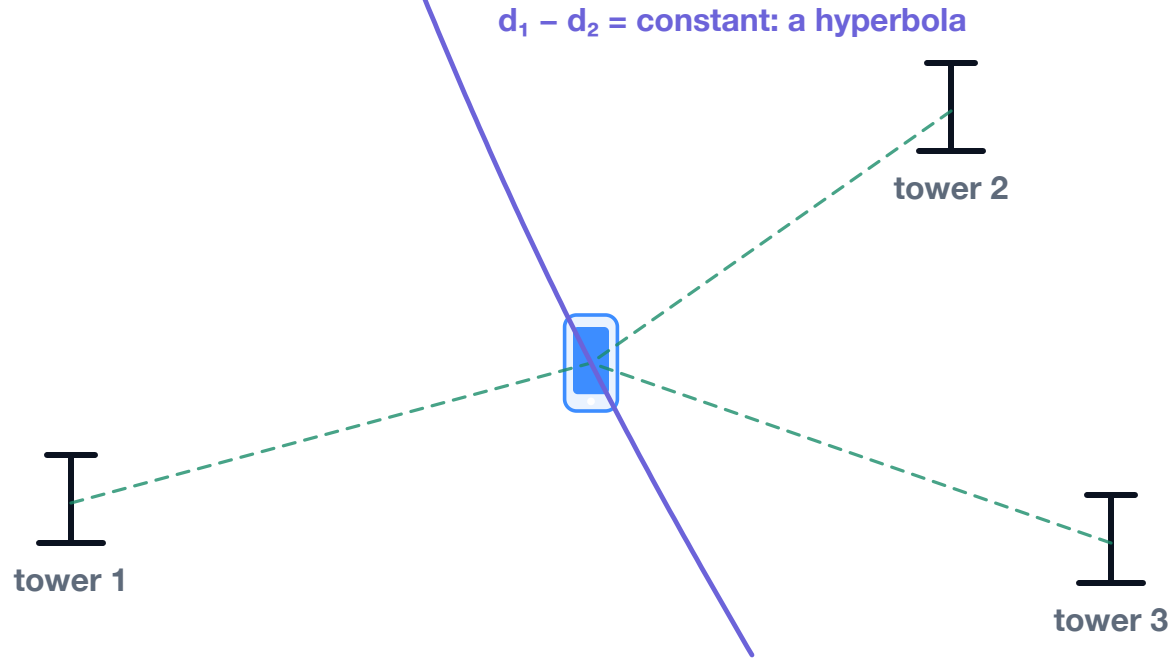
3 Direction

4 Joint

5 Systems

Cellular positioning: the fourth shape

THREE BASE STATIONS, ONE PHONE, NO PHONE CLOCK INVOLVED



the towers share one clock

the phone's clock is not needed: only differences of arrival times are used

a difference of distances

is a hyperbola, the fourth shape on the plan; two of them cross at the phone

5G, release 16 (2020)

reference signals for time differences, round trips and angles at the tower; a few metres indoors is the target

Time difference of arrival. The receiver's clock offset is the same for every tower, so it drops out of every difference.

With synchronised towers, the phone's clock drops out of every difference.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems

Summary

SIX FACTS

- Three things in a received signal say where it came from: **strength, time, direction**.
- Strength falls as $1 / d^2$, but two paths move it by 10 dB in a few centimetres: **room-level, or fingerprint it**.
- Time is distance at 30 cm per nanosecond and does not fade; **two-way ranging removes the clocks**.
- Timing resolution is $1 / B$: 15 m at 20 MHz, 60 cm at 500 MHz. **That is the whole UWB story**.
- Direction is the phase step across the array, read backwards; its width is $\lambda / (N s)$. **Antennas sharpen it**.
- Multipath breaks all three. Measure along more axes, and take the earliest path.

A circle from strength or time, a line from direction, and the earliest path wins.

1 Strength

2 Time

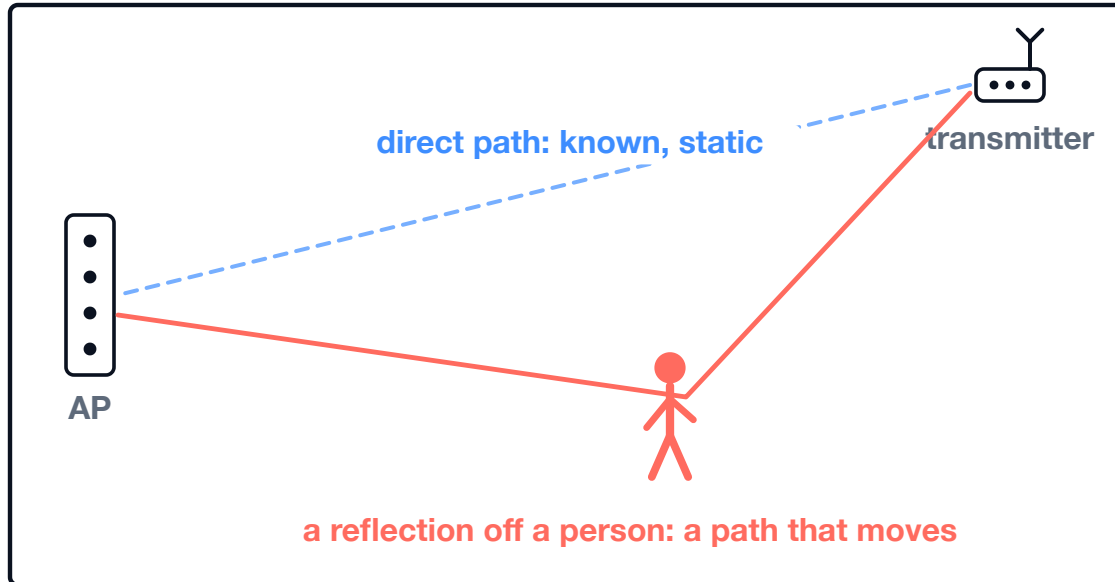
3 Direction

4 Joint

5 Systems

Next: the same signal, and no device to find

THE SAME ANGLE-DELAY PICTURE, WITH NOBODY CARRYING A DEVICE



the person is a path

with an angle, a delay and a Doppler shift of its own

nothing new is measured

the same CSI, the same scan, one more peak

lecture 6: sensing

where a peak is a person, and what it moves like

Localization finds the device that transmits. Sensing finds the body that reflects.

1 Strength

2 Time

3 Direction

4 Joint

5 Systems